APT-BENCH: CLASS-MATCHED SUBJECT–CELL SCALING FOR SINGLE-CELL EVALUATION

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ABSTRACT

Single-cell studies often ask whether a fixed training-cell budget is better allocated to more subjects or more cells per subject. Holding the total number of cells fixed does not isolate this tradeoff: changing subjects can also change the observed labels, per-class counts, and class proportions. We expose this failure mode and introduce a class-matched fixed-total protocol that freezes the training label distribution with within-class donor balancing, together with joint uncertainty that resamples both training subsets and held-out subjects. In label-agnostic experiments, 37 of 38 comparisons across APT, COMBAT, and OneK1K favor 32 rather than 8 training subjects. After class matching, LR and XGBoost effects become small, model- and cohort-dependent, or negative, and all 12 joint intervals include zero; class-matched LR extensions to 64 COMBAT and 128 OneK1K subjects reveal no delayed resolved benefit. APT-Bench supplies the primary patient-disjoint testbed with 361,792 single-cell aptamer profiles from 40 patients, complemented by public cross-cohort evaluations. The central result is therefore methodological: fixed cell totals are insufficient for subject–cell scaling unless label distribution and both sources of sampling uncertainty are made explicit.

1 INTRODUCTION

Single-cell datasets can contain millions of observations while representing far fewer independent biological subjects. This creates a practical design question: under a fixed training-cell budget, should a study sample more subjects or profile more cells from each subject? A common two-axis analysis varies subject breadth and within-subject depth while holding their product fixed. Such a comparison appears controlled because the total number of training cells is unchanged.

Fixed total cells are not sufficient. Replacing cells from existing subjects with cells from new subjects can also introduce previously absent classes, alter per-class cell counts and proportions, and redistribute each class across subjects. A performance difference then combines subject breadth with a change in the training label distribution. Moreover, uncertainty over held-out subjects does not capture sensitivity to which training subjects and cells were sampled. Without controlling both problems, a smooth scaling curve can support the wrong conclusion about the value of additional subjects.

We make this failure mode the central target of APT-Bench. The primary dataset contains 361,792 single-cell aptamer profiles from 40 patients, measured with a fixed 293-feature panel and annotated into five immune lineages and 27 transcriptome-derived subtypes. We first run a conventional label-agnostic fixed-total experiment over subject and cell budgets. We then introduce a class-matched comparison that freezes the observed class set and exact per-class cell quotas before changing the number of contributing subjects. Its primary interval jointly samples a completed training-subset seed and resamples paired held-out subjects, thereby propagating uncertainty from both the development and evaluation populations.

The audit changes the empirical conclusion. Across APT, COMBAT CITE-seq, and OneK1K, 37 of 38 label-agnostic comparisons favor reallocating a fixed cell budget from 8 to 32 subjects. With class quotas fixed and within-class donor balancing, effects become small and model- or cohort-dependent: APT and COMBAT fine-resolution point estimates reverse under both LR and XGBoost, and all 12 joint intervals include zero. Class-matched LR frontiers through 64 COMBAT and 128 OneK1K training subjects show no delayed resolved gain. These results do not imply that subject breadth is unimportant; they show that its effect cannot be read from a fixed-total curve that also changes the prediction problem.

APT hierarchy prediction supplies a concrete cell-annotation setting for this evaluation question; the soft-cascade Transformer is a reference recipe, not the method contribution. Disease characterization, compact panels, and inference-cell budgets remain exploratory appendix audits of the 40-patient cohort. They neither establish clinical utility nor provide independent validation of the scaling protocol.

Our contributions are:

- **A failure mode in subject–cell scaling.** We show that fixed-total, patient-disjoint experiments can still confound subject breadth with observed labels, class counts, and class proportions.
- **A class-matched protocol with joint uncertainty.** The comparison freezes the marginal training label distribution and propagates variation from both sampled training subsets and held-out subjects.
- **Cross-cohort evaluation evidence.** Experiments on APT, COMBAT, and OneK1K show how the matched protocol changes the apparent scaling conclusion across measurement settings; controlled simulations assess a specified allocation target rather than a universal causal effect.

2 Related Work

Evaluation across cells and subjects. Reusable single-cell benchmarks increasingly separate datasets, models, and metrics and adopt sample- or study-disjoint evaluation (Luecken et al., 2025; Liu et al., 2025; Shitov et al., 2025; Liu et al., 2026). These practices prevent direct leakage, but do not by themselves isolate the effect of subject breadth when a fixed-total subsample changes the training label distribution. Our contribution complements patient-disjoint splitting with class-matched allocation and joint development/test uncertainty.

Single-cell representations and hierarchy. Geneformer, scGPT, and scFoundation motivate large transcriptomic representations, while controlled studies show that simple baselines can remain competitive (Theodoris et al., 2023; Cui et al., 2024; Hao et al., 2024; Kedzierska et al., 2025). Cell annotation also motivates hierarchical losses and architectures (Jia et al., 2025; Cultrera di Montesano et al., 2026). We use flat, HCE, and soft-cascade Transformers plus XGBoost as an APT case study. Because their complete objectives are not matched, this ladder illustrates the evaluation setting rather than constituting the primary method or a component ablation.

Clinical assays and measurement budgets. Multi-cancer detection commonly uses circulating DNA, methylation, fragmentomics, or bulk proteins (Cohen et al., 2018; Zeng et al., 2026; Wan et al., 2025; Kahwati et al., 2025). Single-cell APT instead observes many correlated cells from each blood draw, making subject, training-cell, feature, and inference-cell budgets distinct. Gradient-boosted trees are strong tabular baselines (Grinsztajn et al., 2022; Erickson et al., 2026); compact-panel analyses are kept secondary because feature redundancy cannot resolve biological versus technical confounding.

3 Evaluation Setting

3.1 Dataset and Prediction Targets

APT-Bench is derived from peripheral-blood specimens from 40 participants in five cancer groups (CRC, GC, HCC, PC, and BTC) and healthy controls, with 6–8 participants per group. The paired assay measures transcriptomic state and 293 target-specific nucleic-acid aptamers on the same cells. Transcriptomic counts were processed with Seurat (Hao et al., 2021); predicted doublets and low-quality cells were removed, and clusters were manually annotated using canonical markers from prior PBMC and pan-cancer studies (McGinnis et al., 2019; Zhang et al., 2024; Yang et al., 2024). These annotations define the five-lineage and 27-subtype targets. Primary models receive only the

293 APT measurements; the label-generating RNA modality is used only in an internal reconstructability diagnostic. Appendix C.1 gives the complete construction checklist and unresolved provenance fields.

The linked post-QC cohort contains 374,854 cells. Excluding the uncertain *Unknown* subtype leaves 361,792 benchmark cells. Feature transforms are estimated on training patients only. Disease is a six-class patient-level label, whereas lineage and subtype are cell-level labels with one fixed parent lineage per subtype. Disease is therefore audited separately rather than treated as the root of the cell hierarchy. Because each participant contributes one acquisition and acquisition metadata are incomplete, disease analyses estimate within-cohort association, not biological attribution or clinical validity.

TODO: Complete the remaining acquisition, processing, and annotation fields listed in Appendix C.1, including independent review of the subtype-to-lineage lookup.

3.2 Primary Scaling Protocol

The primary contract estimates subject–cell scaling without changing the marginal training label distribution. APT-Bench also fixes patient folds, training-only preprocessing, targets, and metrics for hierarchy and nested patient-characterization case studies. Open-set subtype discovery is outside the contract; the hierarchy uses the fixed five-lineage/27-subtype ontology. The release manifest assigns each pseudonymous patient to one of five disease-stratified folds with eight test patients; all selection remains within development patients. We pool the five out-of-fold (OOF) test partitions. A cell-random split is retained only to quantify leakage. The interface and release requirements are specified in Appendix C.6.

For cell typing, coarse and fine macro-F1 are primary; exact-path accuracy and subtype-tree distance assess the joint hierarchy. Point estimates use pooled OOF cells, while uncertainty uses 2,000 paired patient bootstraps. The prespecified comparison family contrasts the soft-cascade recipe with HCE and XGBoost at both resolutions, with Holm correction over four tests. Subject-balanced sensitivities first normalize each held-out subject's confusion matrix to unit mass before pooling; this differs from both cell-weighted pooling and averaging within-subject macro-F1. Full definitions are in Appendix C.9.

The scaling track varies training subjects $P \in \{8, 16, 32\}$ and cells per subject $C \in \{100, 200, 400, 800, 1600\}$ while keeping outer-test subjects fixed. Fixed-total diagonals hold $T = PC \in \{3200, 6400, 12800\}$ constant, with 20 nested subject/cell subset seeds. The primary metric is subject-balanced pooled OOF macro-F1. Because these reallocations can also change observed classes, per-class counts, and class proportions, a class-matched control freezes the $P = 8$ class set and exact per-class quotas before comparing $P = 8$ with $P = 32$. LR is evaluated at $T \in \{3200, 6400\}$ and XGBoost at $T = 6400$; joint intervals first resample a training-subset seed and then paired test subjects. Appendix Sections C.10–C.12 give allocation, metric, and uncertainty formulas.

We transport the frozen protocol to COMBAT CITE-seq (105 evaluation donors, 700,706 cells; RNA and matched ADT) and OneK1K (925 evaluation donors, 1,199,468 cells; RNA) (COVID-19 Multi-omics Blood ATlas (COMBAT) Consortium, 2022; Yazar et al., 2022). Feature selection uses separate calibration donors. Class-matched LR frontiers extend to 64 COMBAT and 128 OneK1K training donors to test whether the common $P \leq 32$ grid truncates continued improvement. Construction and coverage audits are in Appendix C.11.

The subtype distribution is long-tailed, motivating macro-F1 and patient-clustered uncertainty. Disease association is evaluated at the patient level; cell-level source prediction appears only as a confounding diagnostic. Aptamer and inference-cell budgets are secondary, outer-test-isolated sensitivities and do not imply reduced training-cell requirements.

TODO: Deposit the versioned data, split manifest, and executable evaluator specified in Appendix C.6.

3.3 Case-Study Metrics

All hierarchy models use the same training patients and ontology. Macro-F1 gives each class equal weight despite the long tail; accuracy and root-excluded hierarchical F1 are secondary. Per-subtype support, equal-cell-per-patient sensitivity, and error-conditioned hierarchy diagnostics are reported in Appendix C.8 rather than used as additional primary hypotheses.

4 Evaluation Instantiations

The primary scaling audit uses class-weighted multinomial Logistic Regression and XGBoost to test whether the correction is model-specific. The secondary APT hierarchy case study additionally contains a flat Transformer, the same model trained with a two-level adaptation of HCE (Cultrera di Montesano et al., 2026), and a soft-cascade Transformer. These references share the APT tokenizer, encoder dimensions, optimizer, training budget, patient folds, training-only preprocessing, and validation fine-macro-F1 checkpoint rule. The ladder therefore holds backbone capacity fixed while comparing complete reference recipes. Their objectives are not matched: only the soft-cascade model uses disease labels for cross-disease contrastive pairing, whereas Flat and HCE use classification losses alone. The ladder is therefore not a component ablation and is secondary to the scaling protocol. Appendix B records the implementation and component matrix; no routing, HCE, or contrastive effect is claimed in isolation.

For the composition bridge, each disease outer fold seals eight patients. A final disease-label-free XGBoost subtype model trained on the other 32 patients constructs outer-test compositions. Downstream training compositions for those 32 patients are instead produced by four inner patient-disjoint fits, each trained on 24 patients. A class-weighted multinomial LR, with regularization selected by patient-level inner CV, is then evaluated once on the sealed eight patients. Thus neither the cell-typing stage nor the disease stage uses an outer-test patient during preprocessing, feature construction, fitting, or selection. Appendix B.4 provides the complete data flow and controls. A conditional increment audit uses the same nested features and tuning protocol to compare recorded QC plus predicted lineage against recorded QC plus predicted subtype composition; subtype probabilities implicitly contain their lineage sums.

5 Results

5.1 Fixed Total Cells Do Not Isolate Subject Breadth

We begin with the conventional label-agnostic design: outer-test subjects remain fixed while a constant number of training cells is reallocated from 8 to 32 subjects. Across APT, COMBAT, and OneK1K, 37 of 38 model–resolution–budget comparisons favor greater subject breadth. The direction appears strikingly consistent across assays and cohorts.

It is not an isolated subject effect. Increasing the number of sampled subjects also changes which labels are observed, how the fixed cell budget is distributed among classes, and the resulting class proportions. For example, the mean number of observed fine classes rises with subject count in all three datasets, despite fixed total cells. Training-subset intervals are also much less stable than intervals that condition on the averaged training subsets and bootstrap only test subjects. The naive experiment therefore combines a label-distribution intervention with subject breadth and understates uncertainty from development sampling.

5.2 Class Matching and Joint Resampling Change the Conclusion

The corrected comparison freezes the $P = 8$ observed class set and exact per-class quotas, then samples those same quotas from 8 and 32 subjects. Every joint replicate first selects one completed training-subset seed and then resamples paired outer-test subjects. This targets the residual effect of redistributing a fixed class-matched cell budget across more subjects while propagating both uncertainty sources.

The correction removes the apparent general rule (Figure 1). At $T = 6{,}400$, LR/XGBoost effects are $+0.0078/+0.0083$ for APT coarse but $-0.0050/-0.0058$ for APT fine. COMBAT effects

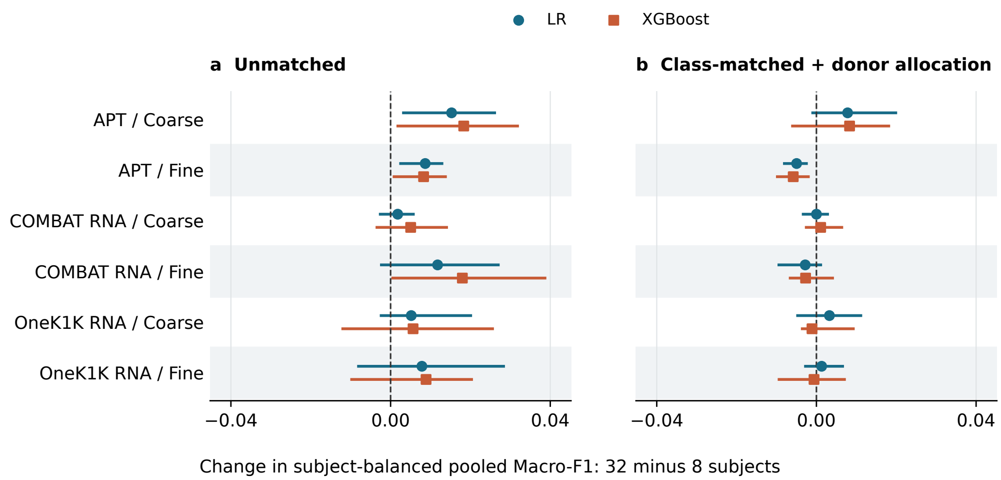

Figure 1: The apparent breadth advantage changes under the matched protocol. Both panels show the change in subject-balanced pooled OOF macro-F1 from 8 to 32 training subjects at $T = 6{,}400$, on identical axes. The right panel fixes class support and exact quotas and additionally uses within-class donor round robin; this is not a label-only intervention. Points are seed means; bars in both panels are empirical 2.5th–97.5th seed percentiles, not confidence intervals of the mean. Unmatched runs and matched LR use 20 seeds; matched XGBoost uses 10. APT and COMBAT fine point estimates change sign for both models. Cross-panel changes are descriptive, not paired attenuation tests. The matched joint intervals, which additionally resample held-out subjects, all span zero (Appendix Figure 4).

are approximately zero for coarse and $-0.0028/-0.0027$ for fine. OneK1K effects are small and change sign across models. All 12 joint intervals include zero. The APT and COMBAT fine reversals are therefore not specific to a linear classifier, but no corrected effect is resolved. This is a joint sampling-protocol contrast: the implementation fixes observed labels, class counts, and proportions and also uses within-class donor round robin. The attenuation cannot therefore be assigned solely to label distribution. An independently controlled synthetic check separates quota matching from donor allocation (Appendix C.4); it does not replace this separation on the real cohorts. A complementary known-null validation yields mean effects $+0.0202$, $+0.0119$, and $+0.0004$ for unmatched, support-matched, and exact-quota sampling against an exactly zero overlap-quota target. Independent Monte Carlo references under donor shifts remain positive and are not mechanically removed by matching (Appendix C.5). This validates a specified allocation estimand, not a universal causal effect of subject breadth.

The result persists beyond the common 32-subject grid. Class-matched LR frontiers through 64 COMBAT and 128 OneK1K training donors yield adjacent effects between -0.00065 and $+0.00123$, with every joint interval spanning zero. Zero inclusion does not establish equivalence or a plateau. A post-hoc ± 0.01 sensitivity bound contains 11 of 12 adjacent frontier joint intervals, but only 4 of 12 main $T = 6{,}400$ intervals and none of the four APT intervals (Appendix C.14). These are pointwise interval-containment diagnostics, not a universal saturation or optimal-allocation result. Complete unmatched grids, class-support audits, matched doublings, extended frontiers, and uncertainty decompositions appear in Appendix C.10– C.13.

5.3 APT Hierarchy as a Case Study

APT cell typing illustrates why both statistical unit and granularity must be part of a scaling contract. Cell-random splitting inflates coarse/fine macro-F1 by $+0.090/+0.108$ for the soft cascade and $+0.050/+0.069$ for XGBoost, whereas equal-cell-per-patient evaluation changes patient-disjoint scores by at most 0.009. Preventing patient overlap, rather than merely reweighting cells, is therefore the consequential control.

Under patient-disjoint evaluation, all models have lower raw fine than coarse F1 (Appendix Table 2). The soft-cascade recipe reaches coarse/fine macro-F1 0.326/0.152 and the flat Transformer 0.323/0.147; no soft-cascade comparison with Flat, HCE, or XGBoost excludes zero. Because only the soft cascade uses cross-disease contrastive regularization, these are complete-recipe comparisons, not evidence for an individual hierarchy component.

A paired-RNA probe reaches 0.790/0.442, versus 0.288/0.104 for APT under the same linear-probe protocol; APT+RNA raises RNA fine macro-F1 by $+0.031$ (95% CI $[0.016, 0.046]$). This establishes an input-dependent reconstruction difference under this probe, not an intrinsic limit of the APT modality. These five-class and 27-class tasks also differ in entropy, imbalance, and label boundaries; their raw F1 difference does not isolate biological resolution. In the matched linear-probe diagnostic, a marginal-chance adjustment reduces the APT coarse-minus-fine difference from 0.184 to 0.051 (paired patient-bootstrap 95% interval $[0.030, 0.080]$; Appendix C.3). The residual is descriptive: chance adjustment does not equate task cardinality or within-lineage difficulty. Per-subtype support and error-conditioned hierarchy analyses remain exploratory and are reported in Appendix C.8.

5.4 Exploratory Patient-Level Audit

The disease audit has 40 independent patients, not 361,792 independent outcomes. Under strict nested cross-fitting, the subtype-minus-lineage patient macro-F1 difference remains unresolved ($+0.116$, 95% patient-bootstrap CI $[-0.015, 0.262]$), with no resolved subtype increment over the recorded nuisance controls. Correct splitting does not distinguish biological from unrecorded acquisition effects. We therefore treat these results as exploratory within-cohort associations, not evidence of incremental subtype utility, cross-task error propagation, or clinical validity. Complete disease, compact-panel, and inference-cell audits appear in Appendix C.7 and Appendix C.17; the scaling contribution does not depend on their predictive performance.

6 Discussion

The central finding is not that more subjects are unimportant, but that a standard fixed-total experiment does not estimate their isolated value. Across three datasets, the label-agnostic design produces an apparently general subject-breadth advantage. Freezing class quotas with within-class donor balancing removes that conclusion: corrected effects are small, model- and cohort-dependent, and unresolved once training-subset and test-subject uncertainty are propagated jointly. Patient-disjoint splitting is necessary, but it is not sufficient for interpreting subject–cell scaling. Zero inclusion does not establish practical equivalence: at a post-hoc ± 0.01 tolerance none of the main APT intervals is fully contained.

The class-matched protocol deliberately targets a narrower estimand. It holds observed labels, per-class cell counts, and class proportions fixed while changing how many subjects contribute those cells. It does not fix disease–subtype composition, within-class subject heterogeneity, or unrecorded pool and batch structure. The remaining effect is therefore still associational, and a near-zero estimate under the tested budgets is not a universal saturation law. In particular, the existing real-data contrast changes both label quotas and within-class allocation, so it does not identify their separate contributions. The synthetic check shows that quota matching need not mechanically erase a breadth benefit, but does not establish real-data mediation. The protocol provides a reproducible sensitivity analysis rather than a unique causal explanation of the observed regularity.

The 40-patient APT cohort also limits patient-level inference: six disease groups contain only 6–8 independent patients each. Neither more cells nor additional resampling supplies independent clinical replication. Nested validation controls information flow, but does not resolve limited precision, adaptive analysis choices, or unmeasured acquisition confounding. The disease audits are consequently exploratory and are not prerequisites for the scaling contribution. COMBAT and OneK1K extend the scaling evaluation, not the clinical validation of APT disease prediction. More generally, subject and cell budgets, label support, split units, and both uncertainty sources should be explicit in single-cell evaluation.

7 Conclusion

Fixed cell totals can confound subject breadth with the training label distribution. Across APT, COMBAT, and OneK1K, class matching with within-class donor balancing changes the apparent advantage, while joint resampling reveals uncertainty hidden by conditioning on training subsets. These comparisons do not identify a unique causal mechanism or establish equivalence between sampling strategies. APT-Bench contributes an explicit evaluation protocol for subject–cell scaling, not a universal scaling law, model-superiority claim, or clinically validated disease predictor.

AI use statement

TODO: **Required for ICLR 2027.** Disclose every generative-AI tool and model used for writing, editing, coding, analysis, literature review, or figure preparation; identify the tasks for which each tool was used, describe human verification of AI-assisted outputs, and state that the authors take responsibility for the final text, claims, code, and artifacts.

Ethics statement

TODO: State the human-subjects approval or exemption and informed-consent status; describe data de-identification, access restrictions, privacy protections, intended clinical/research use, foreseeable misuse or bias risks, and any conflicts of interest. Do not include identifying institutional details if they would break anonymity.

Reproducibility statement

TODO: Provide the anonymous code and data-access locations, exact patient-split manifest, preprocessing and evaluation commands, software environment, random seeds, model hyperparameters, compute requirements, and paths for the prediction artifacts and tables used in every reported result.

Author Contributions

TODO: Add author contributions using CRediT roles (for example: conceptualization, methodology, software, validation, formal analysis, data curation, visualization, writing, supervision, and funding acquisition). Preserve anonymity in the review version if required.

Acknowledgments

TODO: Add funding sources and grant numbers, computational resources, data or clinical contributors, and non-author assistance; verify whether any acknowledgment must be omitted or anonymized during double-blind review.

References

Dvir Aran, Agnieszka P Looney, Leqian Liu, Esther Wu, Valerie Fong, Austin Hsu, Suzanna Chak, Ram P Naikawadi, Paul J Wolters, Adam R Abate, et al. Reference-based analysis of lung single-cell sequencing reveals a transitional profibrotic macrophage. *Nature immunology*, 20(2):163–172, 2019.

Joshua D Cohen, Lu Li, Yuxuan Wang, Christopher Thoburn, Bahman Afsari, Ludmila Danilova, Christopher Douville, Ammar A Javed, Fay Wong, Austin Mattox, et al. Detection and localization of surgically resectable cancers with a multi-analyte blood test. *Science*, 359(6378):926–930, 2018.

COVID-19 Multi-omics Blood ATlas (COMBAT) Consortium. A blood atlas of COVID-19 defines hallmarks of disease severity and specificity. *Cell*, 185(5):916–938.e58, 2022. doi: 10.1016/j.cell.2022.01.012.

Haotian Cui, Chloe Wang, Hassaan Maan, Kuan Pang, Fengning Luo, Nan Duan, and Bo Wang. scgpt: toward building a foundation model for single-cell multi-omics using generative ai. *Nature methods*, 21(8):1470–1480, 2024.

Sebastiano Cultrera di Montesano, Davide D'Ascenzo, Srivatsan Raghavan, Ava P Amini, Peter S Winter, and Lorin Crawford. Improving atlas-scale single-cell annotation models with hierarchical cross-entropy loss. *Nature Computational Science*, 6(3):243–249, 2026.

CZI Cell Science Program, Shibla Abdulla, Brian Aevermann, et al. CZ CELLxGENE Discover: a single-cell data platform for scalable exploration, analysis and modeling of aggregated data. *Nucleic Acids Research*, 53(D1):D886–D900, 2025. doi: 10.1093/nar/gkae1142.

Nick Erickson, Lennart Purucker, Andrej Tschalzev, David Holzmüller, Prateek Desai, David Salinas, and Frank Hutter. Tabarena: A living benchmark for machine learning on tabular data. *Advances in Neural Information Processing Systems*, 38, 2026.

Yury Gorishniy, Ivan Rubachev, Valentin Khrulkov, and Artem Babenko. Revisiting deep learning models for tabular data. In *Advances in Neural Information Processing Systems*, volume 34, pp. 18932–18943, 2021.

Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. Why do tree-based models still outperform deep learning on tabular data? arxiv. *arXiv preprint arXiv:2207.08815*, 2022.

Minsheng Hao, Jing Gong, Xin Zeng, Chiming Liu, Yucheng Guo, Xingyi Cheng, Taifeng Wang, Jianzhu Ma, Xuegong Zhang, and Le Song. Large-scale foundation model on single-cell transcriptomics. *Nature methods*, 21(8):1481–1491, 2024.

Yuhan Hao, Stephanie Hao, Erica Andersen-Nissen, William M Mauck, Shiwei Zheng, Andrew Butler, Maddie J Lee, Aaron J Wilk, Charlotte Darby, Michael Zager, et al. Integrated analysis of multimodal single-cell data. *Cell*, 184(13):3573–3587.e29, 2021. doi: 10.1016/j.cell.2021.04.048.

Shangru Jia, Artem Lysenko, Keith A Boroevich, Alok Sharma, and Tatsuhiko Tsunoda. schdeepinsight: a hierarchical deep learning framework for precise immune cell annotation in single-cell rna-seq data. *Briefings in Bioinformatics*, 26(5):bbaf523, 2025.

Leila C Kahwati, Matthew Avenarius, Leslie Brouwer, Norah L Crossnohere, Chyke A Doubeni, Cecelia Miller, Mariam Siddiqui, Christiane Voisin, Roberta C Wines, and Daniel E Jonas. Multicancer detection tests for screening: a systematic review. *Annals of Internal Medicine*, 178(11): 1591–1604, 2025.

Kasia Z Kedzierska, Lorin Crawford, Ava P Amini, and Alex X Lu. Zero-shot evaluation reveals limitations of single-cell foundation models. *Genome Biology*, 26(1):101, 2025.

Daniel Lakens. Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4):355–362, 2017. doi: 10.1177/1948550617697177.

Chunlei Liu, Sichang Ding, Hani Jieun Kim, Siqu Long, Di Xiao, Shila Ghazanfar, and Pengyi Yang. Multitask benchmarking of single-cell multimodal omics integration methods. *Nature Methods*, 22:2449–2460, 2025. doi: 10.1038/s41592-025-02856-3.

Tianyu Liu, Edward De Brouwer, Archit Verma, Alsu Missarova, Tony Kuo, Nathaniel Diamant, Hanchen Wang, Minsheng Hao, Tommaso Biancalani, Hector Corrada Bravo, et al. Learning multi-cellular representations of single-cell transcriptomics data enables characterization of patient-level disease states. *Cell Systems*, 17(5):101570, 2026. doi: 10.1016/j.cels.2026.101570.

Malte D Luecken, Scott Gigante, Daniel B Burkhardt, Robrecht Cannoodt, Daniel C Strobl, Nikolay S Markov, Luke Zappia, Giovanni Palla, Wesley Lewis, Daniel Dimitrov, et al. Defining and benchmarking open problems in single-cell analysis. *Nature Biotechnology*, 43:1035–1040, 2025. doi: 10.1038/s41587-025-02694-w.

Christopher S McGinnis, Lyndsay M Murrow, and Zev J Gartner. Doubletfinder: Doublet detection in single-cell rna sequencing data using artificial nearest neighbors. *Cell Systems*, 8(4):329–337.e4, 2019. doi: 10.1016/j.cels.2019.03.003.

Vladimir Shitov, Mohammad Moghareh Dehkordi, and Malte D Luecken. Benchmarking sample representations from single-cell data: Metrics for biologically meaningful embeddings. In *ICLR Workshop on Learning Meaningful Representations of Life*, 2025. URL `https://openreview.net/forum?id=IoRv5afWtb`.

Christina V Theodoris, Ling Xiao, Anant Chopra, Mark D Chaffin, Zeina R Al Sayed, Matthew C Hill, Helene Mantineo, Elizabeth M Brydon, Zexian Zeng, X Shirley Liu, et al. Transfer learning enables predictions in network biology. *Nature*, 618(7965):616–624, 2023.

Jonathan CM Wan, Peter Sasieni, and Nitzan Rosenfeld. Promises and pitfalls of multi-cancer early detection using liquid biopsy tests. *Nature Reviews Clinical Oncology*, 22(8):566–580, 2025.

Yu Yang, Xueyan Chen, Jieying Pan, Huiheng Ning, Yaojun Zhang, Yufei Bo, Xianwen Ren, Jiesheng Li, Shishang Qin, Dongfang Wang, et al. Pan-cancer single-cell dissection reveals phenotypically distinct b cell subtypes. *Cell*, 187(17):4790–4811, 2024.

Seyhan Yazar, Jose Alquicira-Hernandez, Kristof Wing, et al. Single-cell eqtl mapping identifies cell type-specific genetic control of autoimmune disease. *Science*, 376(6589):eabf3041, 2022. doi: 10.1126/science.abf3041.

Yong Zeng, Dor D Abelman, Althaf Singhawansa, Nicholas Cheng, Yuanchang Fang, Sasha C Main, Emma Bell, Wenbin Ye, Ping Luo, Samantha L Wilson, et al. A pan-cancer compendium of 1,294 plasma cell-free dna methylomes and fragmentomes enabling multicancer detection. *Nature Cancer*, 7(2):384–398, 2026.

Yuanhang Zhang, Xiaorui Zhou, Yu Zhong, Xi Chen, Zeyu Li, Rui Li, Pengfei Qin, Shanshan Wang, Jianhua Yin, Shang Liu, et al. Pan-cancer scrna-seq analysis reveals immunological and diagnostic significance of the peripheral blood mononuclear cells. *Human Molecular Genetics*, 33(4):342–354, 2024.

A Appendix

B Hierarchy-Aware Reference Ladder

APT-Bench evaluates progressively structured, backbone- and budget-matched neural references: a flat Transformer, its HCE counterpart, and Soft-cascade Transformer. The cascade focuses on the genuine cell-level hierarchy, lineage → subtype, and leaves disease-association auditing to separate tabular baselines and patient-level aggregation. Its released code retains the historical identifier `DropCascade` for artifact compatibility. That identifier denotes a reproducible benchmark baseline, not a proposed algorithm or a claim that its global path, auxiliary coarse objective, soft routing, or contrastive construction independently improves performance.

B.1 Cell Encoder

Each cell is represented by a standardized APT vector $x \in \mathbb{R}^{293}$. We embed feature identity and feature value separately. For aptamer index i, we form a token

$$t_i = \mathrm{Emb}(i) + \mathrm{MLP}_{\mathrm{val}}(x_i), \qquad i = 1, \ldots, 293. \tag{1}$$

A learnable `[CLS]` token is prepended to the resulting sequence and processed by a pre-norm Transformer encoder with 4 layers, 8 attention heads, hidden dimension 256, feed-forward dimension 512, GELU activations, and dropout 0.1. The final `[CLS]` representation

$$z = \mathrm{Encode}(x) \in \mathbb{R}^{256} \tag{2}$$

serves as the shared cell representation for downstream tasks.

B.2 Soft-Cascade Reference

Let $M = 5$ be the number of coarse lineages and $K = 27$ the number of fine subtypes. Each subtype k belongs to exactly one parent lineage $m(k)$. A coarse head produces lineage logits

$$c = W_c z \in \mathbb{R}^M, \tag{3}$$

with routing distribution

$$p(m \mid x) = \mathrm{softmax}(c/\tau)_m, \tag{4}$$

where $\tau = 1$.

Instead of a hard cascade, which would force subtype prediction to follow only the top predicted lineage, Soft-cascade Transformer uses a soft mixture of a global subtype path and lineage-specific experts:

$$s_k(x) = \underbrace{g_k(z)}_{\text{global path}} + \alpha\, p(m(k) \mid x)\, e_{m(k),k}(z) + \gamma \log\big(\epsilon + (1-\epsilon) p(m(k) \mid x)\big), \tag{5}$$

Here $g(z)$ is a global subtype head, $e_{m,k}(z)$ is the expert associated with lineage m, $\alpha = 1.0$, $\epsilon = 0.1$, and γ is ramped from 0 to 0.5 during early training. The global path retains a direct subtype-logit contribution alongside the routed expert contribution; we do not isolate its effect from the other components. Final predictions are

$$\hat{k} = \arg\max_k s_k(x), \qquad \hat{m} = \arg\max_m c_m. \tag{6}$$

B.3 Cross-Disease Contrastive Regularization

The reference model includes a cross-disease supervised contrastive term. Let

$$u = \frac{\mathrm{Proj}(z)}{\|\mathrm{Proj}(z)\|_2} \tag{7}$$

be a normalized projection of the encoder representation. For anchor cell i, positives are cells with the same subtype but a different disease label. Same-subtype cells from the same disease are excluded by construction. Cells from different subtypes are treated as negatives, with additional emphasis on negatives that share the same coarse lineage. The contrastive loss weight is ramped from 0 to 0.2 after the early warmup period. Because we do not compare this construction with standard supervised contrastive learning across all five folds, we make no claim that the cross-disease positive rule independently improves generalization.

B.4 Optimization and Model Scope

The classification cascade is trained jointly using

$$\mathcal{L} = \lambda_{\text{fine}}\mathcal{L}_{\text{fine}} + \lambda_{\text{coarse}}\mathcal{L}_{\text{coarse}} + w_{\text{contrast}}(t)\mathcal{L}_{\text{contrast}}, \tag{8}$$

where $\mathcal{L}_{\text{fine}}$ and $\mathcal{L}_{\text{coarse}}$ are cross-entropy losses with $\lambda_{\text{fine}} = 1.0$ and $\lambda_{\text{coarse}} = 0.5$. The contrastive weight increases to 0.2 according to the schedule in §B.3.

We optimize the cascade using AdamW with learning rate 10^{-4} and weight decay 10^{-5}, gradient clipping at norm 1.0, batch size 512, and mixed-precision training. Models are trained for at most 50 epochs with early stopping based on validation fine-level macro-F1 with patience 10. The final configuration does not use class-balanced sampling because it underperformed natural sampling after matching the number of optimization steps; the single-split development diagnostic is provided in Appendix C.18.

Backbone-matched Transformer references. The flat FT-style Transformer reference (Gorishniy et al., 2021) removes the cascade while retaining the identical APT tokenizer, Transformer dimensions, optimizer, early stopping, and training budget above. We call it FT-style because its tokenizer is exactly the APT feature tokenizer used by Soft-cascade Transformer, rather than the original numerical/categorical FT-Transformer tokenizer. This model and its HCE counterpart predict the same $M + K$ lineage-and-subtype nodes. The flat model applies independent cross-entropy to the lineage and subtype logit slices. For HCE (Cultrera di Montesano et al., 2026), let $R_{ij} = 1$ when node j is node i or its descendant, $p = \mathrm{softmax}(q)$ over all nodes, and $s = Rp$. We minimize $-\lambda_{\mathrm{fine}} \log s_{M+k} - \lambda_{\mathrm{coarse}} \log s_m$ for observed subtype k and lineage m. Because every fine label here is a leaf, the fine HCE term alone would reduce to ordinary CE; including the simultaneously observed parent label makes this a stated two-level adaptation rather than an unqualified reproduction of the mixed-granularity setting in the original work. All three Transformer models are evaluated from the checkpoint selected by patient-validation fine macro-F1, without an additional post-selection calibration stage. Table 1 makes the remaining objective differences explicit. In particular, only the soft-cascade reference uses disease labels to construct cross-disease contrastive pairs; Flat and HCE use no disease labels and no contrastive term. The ladder therefore holds backbone capacity, optimization budget, folds, and checkpoint selection, but it is not an objective-matched component ablation. We do not attribute any score difference to soft routing, hierarchical loss, or contrastive regularization in isolation.

Component	Flat FT-style Transformer	FT-style Transformer + HCE	Soft-cascade Transformer
APT tokenizer and Transformer dimensions	Same	Same	Same
Patient folds, optimizer, budget, checkpoint rule	Same	Same	Same
Fine-label supervision	Independent CE	Leaf HCE NLL	CE
Coarse-label supervision	Independent CE	Ancestor HCE NLL	CE
Cross-disease supervised contrastive term	No	No	Yes ($w \leq 0.2$)
Disease labels used by cell model	No	No	Contrastive pairing only
Global subtype path and lineage experts	No	No	Yes
Soft lineage routing	No	No	Yes

Table 1: Components of the three neural references. "Same" denotes an identical specification, not shared fitted weights. The models match the tokenizer, encoder dimensions, data folds, optimizer, maximum epochs, early stopping rule, and checkpoint criterion, but they do not match the complete training objective. In particular, disease labels enter only the soft-cascade contrastive pairing rule. Consequently, comparisons among these rows evaluate complete recipes and are not component ablations.

Disease recognition is not handled by a dedicated head inside Soft-cascade Transformer. This is a deliberate modeling decision rather than a claim that disease is solved in a strong external sense: within-cohort disease prediction is already nearly saturated by simple classical baselines, while the architectural challenge addressed by Soft-cascade Transformer is the unresolved cell-level hierarchy problem.

Strictly nested cell-output transfer audit. For each disease outer fold f, we train a disease-label-free XGBoost subtype classifier on all 32 outer-development patients and predict the cells of the eight untouched patients in f. Disease-training features require a second cross-fit: for each of the other four fixed folds g, a separate subtype model is trained on the 24 patients outside $f \cup g$ and predicts fold g. We average the resulting 27-class probability vectors within patient to obtain soft subtype composition; soft lineage composition sums subtype probabilities under the frozen subtype-to-lineage map. A standardized, class-weighted multinomial LR is fit to the 32 inner-OOF patient vectors, with $C \in \{0.01, 0.1, 1, 10, 100\}$ selected by patient-level inner CV, and evaluated once on the eight outer-test vectors. Raw proportions are primary; CLR with a prespecified 10^{-6} pseudocount is a sensitivity analysis. All uncertainty and permutation analyses resample or shuffle patients, never cells. To test conditional rather than marginal subtype value, we additionally compare three nested feature sets under the same folds and inner selection: recorded-QC summaries alone, QC plus the five predicted lineage proportions, and QC plus the 27 predicted subtype proportions. The last representation contains lineage implicitly because child probabilities sum to their parents.

C Dataset and Evaluation Details

C.1 Dataset Construction and Documentation Checklist

The available study workflow establishes the high-level sequence used to construct APT-Bench: clinical peripheral-blood specimens from five cancer groups and healthy controls are converted to single-cell suspensions; cell-profile and viability quality control precedes a paired single-cell transcriptome and 293-aptamer assay; separate transcriptome and aptamer-barcode libraries are linked

Model	Coarse M-F1	Fine M-F1	Exact Path	Subtype Tree Dist.↓
Soft-cascade Transformer	0.326	**0.152**	0.253	2.431
Flat FT-style Transformer	0.323	0.147	0.257	2.407
FT-style Transformer + HCE	**0.333**$^{\text{n.s.}}$	0.151$^{\text{n.s.}}$	**0.259**	2.418
XGBoost	0.322$^{\text{n.s.}}$	0.143$^{\text{n.s.}}$	0.248	**2.375**

Table 2: Patient-disjoint 5-fold hierarchy evaluation on cell-weighted pooled OOF predictions from 40 patients. Coarse and fine Macro-F1 are primary. Exact Path requires independent coarse and fine predictions to be correct; Tree Distance is 0/2/4 for correct/sibling/cross-lineage subtype predictions. Prespecified soft-cascade comparisons with HCE and XGBoost use 2,000 paired patient bootstraps and Holm correction across four coarse/fine tests; $^{\text{n.s.}}$ denotes no adjusted significance. Transformer rows match backbone and budget but not the complete objective (Table 1); they are recipe comparisons, not component ablations. Remaining diagnostics and full definitions are in Appendix C.16.

at the cell level; and transcriptome-derived clusters and annotations provide the cell labels. For a benchmark release, this overview must be accompanied by the following provenance and protocol fields.

Transcriptomic preprocessing and cell quality control. The transcriptomic count matrix was processed with Seurat following its standard workflow of normalization, feature selection, dimensionality reduction, neighborhood-graph construction, and clustering (Hao et al., 2021). DoubletFinder was used to identify and remove predicted doublets (McGinnis et al., 2019). Low-quality cells were additionally filtered using per-cell UMI count, detected-gene count, and mitochondrial-transcript fraction. These steps preceded cluster annotation and construction of the linked APT benchmark cohort. TODO: Report the Seurat and DoubletFinder versions, every parameter and random seed, the exact UMI/gene/mitochondrial thresholds, and cell counts before and after each filter.

Manual annotation and reference-label reliability. After Seurat clustering, investigators manually assigned lineage and subtype names by inspecting canonical marker-gene expression reported in prior PBMC and pan-cancer studies (Zhang et al., 2024; Yang et al., 2024). No quantitative annotation confidence, inter-annotator agreement, or external reference-transfer score is available. Accordingly, these labels are operational reference annotations, not error-free biological ground truth, and benchmark scores are conditional on this manual taxonomy. The *Unknown* label was assigned when a cluster did not show a coherent canonical marker pattern for the expected PBMC subtypes; this was a manual reject decision rather than a calibrated confidence threshold. Unknown cells are excluded from the primary 27-subtype benchmark. TODO: Provide the full subtype-by-marker citation table, the number and expertise of annotators, whether they were blinded to disease, the adjudication procedure, and the explicit rule used to assign Unknown.

Cross-disease semantics and hierarchy mapping. The same 27 subtype labels and definitions were applied across all six disease groups; no subtype was renamed or redefined for a particular disease. This gives the labels a consistent operational meaning, but it does not prove that marker expression or biological state is invariant across diseases. Each subtype is assigned to one of five parent lineages by a manually curated lookup. TODO: Document independent validation of the subtype-to-lineage mapping, including reviewers, disagreements, adjudication, and any marker-based checks.

C.2 Paired-RNA Internal Reconstructability Oracle

We use the paired transcriptome as a diagnostic oracle to distinguish an APT information limitation from complete non-reconstructability of the operational labels. This analysis is deliberately outside the primary APT-only benchmark. The RNA representation contains 2,000 log-normalized highly variable genes selected by Seurat-style dispersion during data development. All 361,792 benchmark cells match uniquely to this matrix by cell barcode. Because this feature representation and the manual labels arise from the same transcriptomic pipeline, the oracle is circular with respect to label construction and cannot serve as independent validation or a leakage-free model comparison.

Within each of the five fixed patient-disjoint folds, we compare APT only, paired RNA only, and their feature concatenation. The three inputs use the identical deterministic sample of up to 2,000 training cells per development patient and all cells from the eight untouched test patients. Features are standardized using training-fold statistics. We fit the same 30-epoch patient- and class-weighted linear SGD probe at both resolutions; the fixed optimization budget avoids modality-specific tuning. Scores use subject-normalized OOF confusion matrices, and 2,000 paired patient-bootstrap replicates quantify modality contrasts. This matched-budget probe is not directly comparable to the full-training model ladder in Table 2.

Input	Coarse Macro-F1	Fine Macro-F1	Granularity gap
APT only	0.288 [0.267, 0.308]	0.104 [0.086, 0.115]	0.184
Paired RNA only	0.790 [0.775, 0.801]	0.442 [0.399, 0.467]	0.348
APT + paired RNA	0.794 [0.777, 0.807]	0.473 [0.428, 0.499]	0.321

Table 3: Paired-RNA internal reconstructability oracle under the five fixed patient-disjoint folds. All inputs use the same patient- and class-weighted linear probe, up to 2,000 identically sampled training cells per development patient, training-fold standardization, and all held-out cells. Scores are subject-balanced pooled Macro-F1; brackets are patient-bootstrap 95% intervals, and the granularity gap is coarse minus fine Macro-F1. This diagnostic uses the transcriptomic modality from which the annotations were constructed and a fixed 2,000-HVG representation selected during data development, so it is neither independent label validation nor a leakage-free benchmark baseline.

Paired RNA improves coarse/fine macro-F1 over APT by $+0.502$ (95% CI $[0.478, 0.527]$) and $+0.338$ ($[0.300, 0.365]$). The concatenated representation improves fine macro-F1 over RNA alone by $+0.031$ ($[0.016, 0.046]$), whereas its coarse change of $+0.004$ ($[-0.001, 0.009]$) is unresolved. RNA therefore reconstructs the operational taxonomy far better than APT, but its own coarse–fine gap remains 0.348 and the complementary APT fine signal is nonzero. The experiment supports a mixed interpretation involving modality resolution and the difficulty of the patient-disjoint taxonomy; it does not estimate annotation accuracy.

- TODO: Report recruiting site(s) and dates, prospective versus retrospective collection, exact patient count per group, inclusion/exclusion and diagnostic criteria, and whether each participant contributed exactly one specimen.
- TODO: Report available demographics and clinical covariates, including age, sex, disease stage, treatment status, and comorbidities; summarize missingness and state which fields cannot be released.
- TODO: Report blood volume, collection tubes, time to processing, cell-isolation and cryopreservation procedures, quality-control thresholds, and sample/cell exclusion counts.
- TODO: Report assay/platform and reagent versions, sequencing instruments and depth, 293-probe panel design, and a verified protein-target manifest or explicitly state why only anonymized probe identifiers can be released.
- TODO: Report barcode-matching rates, APT normalization and filtering, all remaining software versions and parameters, and whether annotation was performed before investigators examined APT profiles or disease labels.
- TODO: Add a cohort-flow diagram with participant and cell counts after every exclusion step, and provide the ethics approval/consent, de-identification, data-access, license, and intended-use terms referenced in the Ethics and Reproducibility Statements.

C.3 Chance-Adjusted Resolution Diagnostic

Raw coarse and fine Macro-F1 compare five-class and 27-class tasks with different label frequencies and decision boundaries. We therefore treat their difference as descriptive, not evidence of a resolution deficit beyond task cardinality. Using the released 40 patient-specific OOF confusion matrices from the matched linear-probe RNA diagnostic, we normalize each matrix to unit mass and average over patients to obtain $\bar{M}$. Let $p_k = \sum_j \bar{M}_{kj}$ and $q_k = \sum_i \bar{M}_{ik}$. The independence reference

$M^0_{ij} = p_i q_j$ retains both true and predicted marginals while breaking their association. We report

$$S_0 = \frac{1}{K} \sum_k \frac{2p_k q_k}{p_k + q_k}, \qquad S_{\text{adj}} = \frac{\text{MacroF1}(M) - S_0}{1 - S_0}, \tag{9}$$

with zero contribution when $p_k + q_k = 0$. This is F1 of an independence matrix, not the exact expectation of finite-sample permutation F1. Each of 2,000 paired patient-bootstrap draws recomputes M, both marginals, and the adjustment; the same patient draw is used for both resolutions and all modalities. These intervals condition on the fitted models and do not include retraining variation.

Input	Resolution	Raw F1	Chance reference	Adjusted F1 [95% interval]
APT	coarse	0.288	0.186	0.125 [0.103, 0.147]
APT	fine	0.104	0.033	0.074 [0.056, 0.085]
RNA	coarse	0.790	0.198	0.738 [0.719, 0.752]
RNA	fine	0.442	0.035	0.422 [0.374, 0.446]
APT + RNA	coarse	0.794	0.199	0.743 [0.722, 0.760]
APT + RNA	fine	0.473	0.035	0.454 [0.403, 0.481]

Table 4: Marginal-chance adjustment for the matched linear-probe RNA diagnostic. Scores use subject-balanced pooled confusion matrices from the same 40 OOF patients. Each of 2,000 paired patient-bootstrap draws recomputes the observed score and both null marginals. This is an association-breaking reference, not a task-cardinality or annotation-quality control.

For APT, the raw coarse/fine scores 0.288/0.104 have chance references 0.186/0.033 and adjusted scores 0.125/0.074. The difference shrinks from 0.184 to 0.051 (95% interval $[0.030, 0.080]$). Adjusted RNA and APT+RNA differences are 0.316 ($[0.293, 0.356]$) and 0.289 ($[0.264, 0.329]$), respectively. The subject-balanced true-label entropies are 1.253 and 2.488 nats, corresponding to effective class counts $\exp(H)$ of 3.50 and 12.03. These describe rather than match class imbalance. The adjustment accounts for marginal chance agreement, but does not control label boundaries, class count, annotation reliability, or within-lineage separability; none of these residual differences establishes a cardinality-independent biological granularity effect.

True-lineage-conditioned subtype decoding is not evaluated here. It requires OOF subtype scores restricted to the children of the true lineage (or separately trained conditional classifiers); hard predictions and confusion matrices cannot recover that decision. In particular, filtering to predictions that already match the true lineage would select easier cases rather than provide this control. Thus the present diagnostic cannot attribute fine errors uniquely to lineage identification versus within-lineage discrimination.

C.4 Sequential Controls: Synthetic Mechanism Check

The real-data class-matched sampler fixes class quotas and spreads cells across donors within each class. To distinguish these operations, we implement four conditions with common random cell priorities and nested development subject sets: (A) uniform pooled-cell sampling; (B) restriction to the eight-subject reference class set, reserving one cell per class before uniform filling; (C) exact reference class quotas with uniform within-class sampling; and (D) the same quotas with capacity-constrained donor round robin in randomized donor order. At fixed total, exact class proportions and exact counts are equivalent constraints, not separate ablations. This A condition is not the legacy equal-cell-per-subject sampler. Sequential contrasts are order-dependent descriptions, not causal mediation estimates.

We evaluate this implementation on synthetic data, not a semi-synthetic clinical cohort. Each of 20 seeds generates six class centers $\mu_k \sim \mathcal{N}(0, s^2 I_{12})$ and 48 subjects with 400 cells each. Subject label probabilities are $\text{softmax}(\eta_p)$, where $\eta_p \sim \mathcal{N}(0, m^2 I_6)$, and features are $x_{pi} = \mu_{y_{pi}} + b_p + \epsilon_{pi}$, with $b_p \sim \mathcal{N}(0, a^2 I_{12})$ and $\epsilon_{pi} \sim \mathcal{N}(0, I_{12})$ independently. We vary $a, m \in \{0, 1.5\}$ and $s \in \{0.5, 1.5\}$. The first 32 subjects form the development pool; the remaining 16 are untouched test subjects. An unweighted LR probe is refit with training-only standardization for each condition and $P \in \{8, 32\}$ at total 1,600 training cells. All conditions reuse the same realization and test subjects within a seed.

Offset	Mixture	Separation	A: Unmatched	C: Quota	D: Quota + donor
0.0	0.0	0.5	+0.001 [-0.007,+0.007]	+0.001 [-0.007,+0.008]	+0.001 [-0.006,+0.007]
0.0	0.0	1.5	-0.000 [-0.003,+0.001]	-0.000 [-0.003,+0.002]	-0.000 [-0.003,+0.001]
0.0	1.5	0.5	+0.024 [-0.005,+0.075]	-0.001 [-0.012,+0.010]	-0.002 [-0.012,+0.007]
0.0	1.5	1.5	+0.002 [-0.005,+0.009]	+0.000 [-0.005,+0.003]	-0.000 [-0.003,+0.005]
1.5	0.0	0.5	+0.028 [-0.028,+0.103]	+0.028 [-0.023,+0.103]	+0.031 [-0.015,+0.104]
1.5	0.0	1.5	+0.043 [-0.005,+0.114]	+0.042 [-0.006,+0.113]	+0.042 [-0.009,+0.115]
1.5	1.5	0.5	+0.027 [-0.051,+0.133]	+0.016 [-0.059,+0.115]	+0.025 [-0.039,+0.103]
1.5	1.5	1.5	+0.040 [-0.029,+0.130]	+0.029 [-0.054,+0.114]	+0.048 [-0.008,+0.136]

Table 5: Synthetic mechanism check: subject-balanced Macro-F1 difference between 32 and 8 training subjects at fixed total 1,600 cells. Entries are means across 20 independent seeds with empirical 2.5th–97.5th seed percentiles, not confidence intervals of the mean. Condition B equals A in every run because support is already complete and the support floor does not alter selection. All eight parameter settings are reported.

With exchangeable subjects ($a = m = 0$), mean effects are near zero. With label mixture variation alone and $s = 0.5$, the unmatched mean effect is $+0.0245$, compared with -0.0011 after quota matching. Conversely, with subject offsets alone and $s = 0.5$, quota matching retains a mean effect of $+0.0280$, compared with unmatched $+0.0281$. Thus matching does not invariably eliminate a breadth benefit in this generator. Seed variability is substantial; these results do not establish interval coverage, a universal correction, or the mechanism in APT, COMBAT, or OneK1K. Support is complete in these simulations, so they do not test missing-class recovery. Separate real-cohort A–D results are not included in this version; the existing class-matched tables must not be interpreted as that decomposition.

C.5 Known-Null and Independent-Target Validation

Known generator parameters are not themselves a known learning effect. We therefore define the target explicitly as

$$\theta_C = \mathbb{E}[S_C(32, Q_8) - S_C(8, Q_8)]\,, \tag{10}$$

where S is subject-balanced pooled six-subtype Macro-F1 on untouched test subjects, and Q_8 is the development-only eight-subject class-quota reference. The expectation integrates over generated cohorts and sampling. This overlap-quota policy target excludes gains from acquiring labels absent at $P = 8$; it is not the total practical value of recruiting more subjects.

We generate a hierarchy of three lineages with two sibling subtypes each in eight dimensions. Fixed class centers are $\mu_k = 2e_{\lfloor k/2 \rfloor} + (-1)^{k+1} s e_{3+\lfloor k/2 \rfloor}$ for $k = 0, \ldots, 5$ (zero-indexed axes). Features follow $x_{pi} = \mu_{y_{pi}} + b_p + u_{p,g(i)} + v_{h(p)} + \epsilon_{pi}$, with independent isotropic Gaussian effects: donor standard deviation a, ten-cell block standard deviation c, four-donor batch standard deviation b, and residual standard deviation 1. Within-subject correlation can thus arise at donor or cell-block level; these covariance components should not be interpreted as independent empirical estimates of biological heterogeneity. Subject label probabilities are softmax-normal with logit standard deviation 1.5, and the sixth subtype is available in only 10% of subjects. Each cohort contains 32 development and 40 test subjects, with 120 cells per subject. The training total is 480 at both $P = 8$ and $P = 32$; preprocessing and LR fitting follow the preceding synthetic probe. Train and test subjects and batches are disjoint. All tasks retain the fixed six-class scoring ontology.

The base setting is $(a, c, b, s) = (0, 0, 0, 0.8)$. Conditional on the label-only quota reference, training features in each class are iid with the same distribution for both subject budgets. Consequently, the C fitted-model distribution, and hence expected test score, is identical: $\theta_C = 0$ exactly. This argument does not require balanced natural class proportions or complete support. Additional settings change only a to 0.5 or 1.5; stress settings start from $a = 0.5$ and set $c = 0.7$, $s = 0.3$, or $b = 1$, respectively. In the batch setting, increased subject breadth also increases batch coverage; the target remains a policy contrast rather than a separate donor effect.

For every non-null setting we estimate θ_C using 512 independent reference cohorts, with a direct uniform-without-replacement index sampler separate from the evaluated priority-sampling implementation. Another 100 independent cohorts evaluate A/B/C using paired subject pools, cell priorities, and test sets. Reference and evaluation random-seed ranges are disjoint. We report reference Monte Carlo standard errors (MCSE), estimator bias, and RMSE. Normal Monte Carlo intervals for bias combine evaluation-mean variance and independent reference-mean variance; these are not patient-bootstrap coverage measurements or multiplicity-adjusted tests.

Setting	Target (MCSE)	A: mean / RMSE	B: mean / RMSE	C: mean / RMSE
Known null	0 (exact)	+0.0202 / 0.0354	+0.0119 / 0.0249	+0.0004 / 0.0109
Mild donor shift	0.0110 (0.0009)	+0.0278 / 0.0341	+0.0221 / 0.0264	+0.0102 / 0.0191
Strong donor shift	0.0135 (0.0015)	+0.0332 / 0.0412	+0.0316 / 0.0392	+0.0168 / 0.0358
Correlated cells	0.0108 (0.0008)	+0.0301 / 0.0334	+0.0264 / 0.0290	+0.0102 / 0.0199
Weak subtypes	0.0049 (0.0010)	+0.0247 / 0.0361	+0.0235 / 0.0346	+0.0016 / 0.0222
Batch shift	0.0181 (0.0017)	+0.0318 / 0.0463	+0.0299 / 0.0438	+0.0192 / 0.0429

Table 6: Known-null and independent Monte Carlo validation of the overlap-quota target. A, B, and C denote unmatched, support-matched, and exact-quota sampling. Non-null targets use 512 independent reference cohorts; estimator means and RMSE use 100 separate evaluation cohorts. MCSE denotes reference Monte Carlo standard error. The target is the expected effect of the C allocation policy, not unrestricted subject recruitment. All six settings are reported; RMSE is measured relative to the exact null or estimated reference target.

Under the exact null, A and B yield mean effects $+0.0202$ and $+0.0119$, whereas C yields $+0.0004$ (bias interval $[-0.0017, 0.0026]$). The independent reference check is $+0.00043$ with MCSE 0.00053, consistent with the analytic null. Mild and strong donor shifts have reference effects $+0.0110$ and $+0.0135$; C estimates $+0.0102$ and $+0.0168$, respectively, rather than mechanically forcing both to zero. Higher shift variance does not guarantee a proportionally larger learning benefit. Across all six settings, C has lower RMSE against this target than A or B. Nevertheless, A/B differences can be legitimate benefits of their own allocation policies: their error relative to θ_C demonstrates estimand mismatch, not that all observed breadth gains are spurious. Non-null targets remain Monte Carlo approximations. This validation neither identifies real-cohort causal mediation nor establishes coverage of the joint uncertainty procedure used on real data.

C.6 Benchmark Interface and Release Contract

The hierarchy submission interface contains pseudonymous cell and patient IDs, true and predicted lineage, and true and predicted subtype. The disease interface contains one row per patient with the true disease and six class scores. The efficiency track additionally records the aptamer measurement budget, inference-time cell budget, selection seed, and the development-only feature ranking. These common interfaces permit new models to be scored without reproducing their internal training pipelines.

The evaluation layer is portable to other clinical single-cell datasets. It implements group-disjoint out-of-fold prediction; patient-clustered bootstrap uncertainty; coarse and fine macro-F1, exact-path accuracy, hierarchical F1, subtype-tree distance, granularity gap, and sibling versus cross-lineage error; patient-level aggregation versus cell-level pseudoreplication; patient-centering and identity diagnostics; and nested feature- and cell-budget curves. A future dataset can replace the assay features, hierarchy, or clinical endpoint while retaining these interfaces. This modular task–dataset–metric separation follows recent living single-cell benchmarks (Luecken et al., 2025); standardized metadata and programmatic access similarly enable cross-study reuse in CZ CELLxGENE (CZI Cell Science Program et al., 2025).

A complete release must contain the processed APT matrix; pseudonymous cell and patient identifiers; disease and cell labels; hierarchy and feature metadata; the immutable fold manifest; metric implementations; baseline configurations; and a single command that regenerates the headline tables

(a) Task and evaluation scope

Resource	Primary input	Reported scope	Main unit	Hierarchy
Open Problems	Multiple	12 living tasks	Task-specific	Task-specific
scMultiBench	RNA/ADT/ATAC	64 real + 22 simulated	Cell/dataset	No
SPARE	scRNA-seq	5 datasets, 8 methods	Sample	No
PaSCient	scRNA-seq	12.5M cells, 2,700 patients	Patient	No
APT-Bench	293 aptamers	1 cohort, 40 patients	Cell + patient	Yes

(b) Protocol components and availability

Resource	Patient task	Two-axis scale	Assay budget	Public pipeline
Open Problems	Task-specific	No	No	Yes
scMultiBench	No	No	Feature selection	Yes
SPARE	Yes	No	No	Yes
PaSCient	Yes	No	No	Not audited
APT-Bench	Audit	Yes	Yes	Pending release

Table 7: Scope comparison with recent single-cell evaluation resources (Luecken et al., 2025; Liu et al., 2025; Shitov et al., 2025; Liu et al., 2026). “Task-specific” means that the living platform defines multiple contracts rather than one shared clinical protocol. APT-Bench is narrower in cohorts but combines a fixed aptamer modality, patient-disjoint coarse-to-fine cell typing, within-cohort patient-level audit, and controlled patient/cell and assay budgets. Its public release remains a submission requirement, not a completed claim.

from model predictions. TODO: Before submission, add data and code URLs, explicit licenses, file checksums, an environment lockfile, an archival DOI, and the end-to-end command. If participant-level data require controlled access, release the complete schema, all labels permitted by consent, the split manifest, evaluator, and access procedure.

C.7 Confounding Audit of Within-Cohort Disease Association

Exploratory scope and independent sample size. All disease analyses reuse one cohort of 40 independent patients across six groups (6–8 patients per group); the 361,792 cells are repeated observations, not additional independent disease outcomes. Outer-fold disease models are developed on only 32 patients. Nested cross-fitting separates training from evaluation but does not increase the number of independent subjects, remove unmeasured confounding, or turn repeated analyses into independent replications. Regularization and inner validation mitigate model-selection risk without eliminating the uncertainty associated with this small development sample. The representation comparisons and post-hoc sensitivities below are exploratory, not confirmatory evidence of incremental subtype value or clinical utility. Reported patient-bootstrap intervals do not account for the full adaptive history of analysis choices. Clinical validation would require an independent APT cohort with recorded acquisition covariates and a frozen preprocessing, feature-selection, and prediction pipeline; the external scaling experiments do not supply that disease-validation cohort.

Aggregation / Representation	Patient Acc.	Macro-F1	Macro-AUROC	Perm. p
Log cell count + LR[‡]	0.575	0.531	0.800	0.00010
All recorded QC proxies + LR[¶]	0.550	0.566	0.859	–
Nested soft subtype composition + LR[§]	0.600	0.593	0.854	0.00010
Nested soft lineage composition + LR[§]	0.475	0.477	0.766	0.00020
Gold subtype composition + LR[‡]	0.500	0.516	0.776	0.00010
Gold lineage composition + LR[‡]	0.325	0.301	0.743	0.0184

[‡] Patient-level technical or paired-transcriptomic oracle features. [¶] Recovered cell count plus five patient summaries of recorded RNA UMI and detected-gene QC fields; see Appendix C.7. [§] Soft probabilities from disease-label-free XGBoost under strict nested cross-fitting. Composition rows use raw proportions; LR regularization is tuned only within each outer-development set. Composition permutation tests use 10,000 patient-label shuffles.

Table 8: **Associational patient-characterization audit** ($n = 40$). Predicted composition retains disease-associated variation, but its gains over lineage composition, recovered cell count, and recorded RNA-QC proxies have patient-bootstrap intervals spanning zero. These representations may contain biological, patient, or technical structure. Direct APT classifiers and their confounding controls are confined to Appendix C.7; no row establishes biological mediation or clinical generalization.

Cell-level prediction of the disease source of a held-out patient. Table 8 reports the exploratory patient-characterization audit. For completeness, Table 10 reports the pooled *cell-level* source-prediction metric, in which every held-out cell is treated as a target and pooled across all 361,792 cells. This is an easier and statistically different task than patient-level characterization, since cells from the same patient are correlated and patients with more cells contribute more weight. We report it only as a confounding diagnostic.

Non-identifiability of disease and acquisition effects. Every participant contributes one acquisition, so patient identity is nested within both disease group and any unrecorded processing batch. The released metadata contain disease and patient identifiers plus transcriptomic UMI and detected-gene counts, but no plate, run, date, library, operator, or site field. Consequently, neither patient-disjoint prediction nor label permutation can identify whether a patient-associated APT offset is biological, technical, or mixed. No analysis below is interpreted as disease biology, biomarker validation, or clinical performance.

Recorded nuisance-only baselines. We aggregate the only available per-cell QC fields to each patient: recovered cell count; medians of $\log(1 + \mathrm{UMI})$ and $\log(1 + \mathrm{detected\ genes})$; median detected-gene/UMI ratio; and the interquartile ranges of the two log-count variables. Each feature set uses the same outer folds as the composition bridge, four-fold inner patient CV on the 32 outer-development patients to select LR regularization, refitting on all 32, and one evaluation on the sealed eight patients. Table 9 shows that these recorded QC proxies alone approach the predicted-composition result. They remain proxies rather than measured batch labels, so this analysis detects known nuisance structure but cannot rule out unrecorded acquisition effects.

We also test conditional subtype value rather than comparing each representation in isolation. Under the same nested folds and inner regularization selection, recorded QC plus predicted lineage reaches macro-F1 0.583 and AUROC 0.875; recorded QC plus the full predicted subtype vector reaches 0.599 and 0.825. The subtype-minus-lineage macro-F1 increment is $+0.016$ with 95% paired patient-bootstrap CI $[-0.120, 0.169]$ (10,000 resamples). Thus subtype detail does not provide a resolved conditional increment in this cohort.

Recorded feature set	d	**Acc.**	**Macro-F1**	**Macro-AUROC**
Recovered cell count	1	0.575	0.531	0.800
RNA QC summaries	3	0.475	0.478	0.787
All recorded QC proxies	6	0.550	0.566	0.859
Predicted soft subtype composition	27	0.600	0.593	0.854
Recorded QC + predicted lineage	11	0.575	0.583	0.875
Recorded QC + predicted subtype	33	0.600	0.599	0.825

Table 9: Matched patient-level nuisance audit. All rows use the same outer folds and inner patient-level regularization selection. The predicted-composition minus all-recorded-QC Macro-F1 difference is $+0.027$ with a 95% paired patient-bootstrap interval of $[-0.157, 0.215]$ (2,000 resamples). Recorded QC fields are incomplete proxies, not batch labels. In the conditional audit, the subtype-minus-lineage Macro-F1 increment is $+0.016$ with 95% paired patient-bootstrap interval $[-0.120, 0.169]$ (10,000 resamples).

Model	**Accuracy**	**Macro-F1**	**Weighted-F1**	**Balanced Acc.**
Logistic Regression	**0.991**	**0.989**	**0.991**	**0.989**
XGBoost	0.990	0.988	0.990	0.987
Linear SVM	0.988	0.986	0.988	0.985
Random Forest	0.963	0.955	0.963	0.953
Reference Correlation†	0.921	0.909	0.921	0.915

† SingleR-style reference-correlation classifier (Spearman nearest-centroid mapping), following the template-matching principle used by reference-mapping methods such as SingleR (Aran et al., 2019).

Table 10: **Cell-level** prediction of the disease source of a held-out patient's cells (6-class), pooled over all 361,792 out-of-fold cells from the patient-level 5-fold protocol. This is a different, easier statistical task than the patient-level diagnosis reported in Table 8: cells from the same patient are correlated observations rather than independent draws, and this metric implicitly weights patients with more cells more heavily. We report it here only as a cell-level reference point, not as a patient-level diagnostic result.

Patient-wise centering sensitivity. To test whether disease prediction relies on each patient's absolute expression level rather than only on within-patient relative structure, we repeated Logistic Regression training and evaluation after subtracting each patient's own per-aptamer median from every cell in both training and test data. This transformation preserves within-patient cell-to-cell variation while removing a patient-specific location shift. Cell-level accuracy drops from 0.991 to 0.401 (macro-F1 $0.989 \rightarrow 0.348$), and patient-level majority-vote accuracy drops from $40/40$ to $26/40$. This collapse shows that disease recognition depends substantially on patient-level absolute-expression structure, but it does not by itself determine whether that structure is biological, technical, or a mixture of both.

Patient-fingerprint diagnostic. We additionally trained a 40-way classifier to predict patient identity from individual cells, using a within-patient 50/50 cell split for this diagnostic only. The classifier identifies the correct patient from a single held-out cell with 88.4% accuracy (macro-F1 0.862), against a 2.5% chance level. Individual cells therefore carry a strong and easily recoverable patient-specific signature, consistent with the centering sensitivity above.

Patient-level summary-feature control. Finally, to confirm that the disease signal does not rely only on cell-level pseudo-replication, we trained a genuinely patient-level classifier using each patient's median 293-dimensional APT profile. This summary-feature model reaches patient accuracy 0.950, patient macro-F1 0.948, and macro-AUROC 1.000. Together with the sensitivity analyses above, this indicates that disease information is present at the patient level but strongly entangled with absolute-expression offsets that cannot be cleanly separated without acquisition-batch metadata.

Strict nested cell-output transfer audit. Ordinary pooled OOF predictions are insufficient for stacking because a base model used to create a disease-training feature may have seen the eventual disease outer-test patients. We therefore use the nested protocol defined in Methods: within each outer fold, four XGBoost models trained on 24 patients generate inner-OOF features for the 32 disease-training patients, and a fifth model trained on all 32 predicts the eight outer-test patients. XGBoost receives subtype labels but no disease labels. Soft subtype composition reaches accuracy/macro-F1/AUROC $0.600/0.593/0.854$ and soft lineage composition reaches $0.475/0.477/0.766$. The paired macro-F1 difference is $+0.116$ (95% patient-bootstrap CI $[-0.015, 0.262]$). Gold subtype and lineage compositions reach macro-F1 0.516 and 0.301, respectively. The apparently stronger predicted than gold point estimate is not statistically resolved and warns that predicted probabilities may encode model-confidence or technical structure beyond true cell abundance.

This experiment is therefore a representation-transfer audit, not an error- propagation analysis. It does not establish that the transferred signal originates from subtype abundance, that fine subtype is more useful than coarse lineage, that better typing improves patient characterization, or that the signal is independent of confidence and technical structure. Consistent with this distinction, the soft-minus-hard subtype macro-F1 difference is only $+0.069$ (95% patient-bootstrap CI $[-0.077, 0.221]$), and the predicted-minus-gold subtype difference is $+0.077$ (95% CI $[-0.100, 0.256]$).

Across 10,000 complete patient-label permutations, predicted subtype and lineage composition have plus-one $p = 0.00010$ and $p = 0.00020$. However, log cell count alone reaches macro-F1 0.531 ($p = 0.00010$), and the predicted-subtype minus cell-count difference is $+0.062$ (95% CI $[-0.087, 0.218]$). Equal-cell subsampling at 100, 500, and 2,000 cells per patient preserves the qualitative result over 50 seeds. A 50-repeat global-multinomial composition control averages macro-F1 0.170, and fold-ID prediction from global OOF composition is below chance. These controls support a predictive and associational bridge, not causal mediation or independent biological validation.

Representation	Legacy pooled OOF	Strict nested
Gold subtype composition	0.572	0.516
Predicted subtype composition	0.400	0.593
Gold lineage composition	0.377	0.301
Predicted lineage composition	0.282	0.477

Table 11: Protocol audit of patient macro-F1. The legacy predicted rows use hard labels from one global pooled-OOF prediction set, CLR features, and fixed LR regularization; the strict rows use nested soft probabilities, raw proportions, and development-only LR selection. Because both feature construction and downstream fitting changed, the numerical differences are not estimates of a leakage effect. Only strict nested results support current claims.

Patient-label permutation control. We evaluated the same patient-level median-profile classifier under 1,000 label permutations. Disease labels were permuted across the 40 patients as indivisible units, preserving the disease-group counts $(6, 8, 7, 7, 6, 6)$; labels were never shuffled across cells. We retained the original five patient folds rather than restratifying after permutation. No permuted dataset reached the observed accuracy of 0.950 or macro-F1 of 0.948, giving plus-one permutation $p = (0 + 1)/(1000 + 1) = 0.000999$ for both statistics. The null accuracy mean was 0.137, with empirical 95% interval $[0.025, 0.275]$. This rejects a random patient-label explanation under the fixed protocol, but does not distinguish biological signal from patient-associated technical structure.

C.8 Per-Subtype Support and Performance Audit

Figure 2 visualizes the two marginal support–F1 associations for Soft-cascade Transformer and XGBoost. The vertical stack at 40 patients makes the coverage ceiling explicit. These plots and the partial correlations are exploratory: they neither establish causality nor test whether two dependent correlations differ significantly.

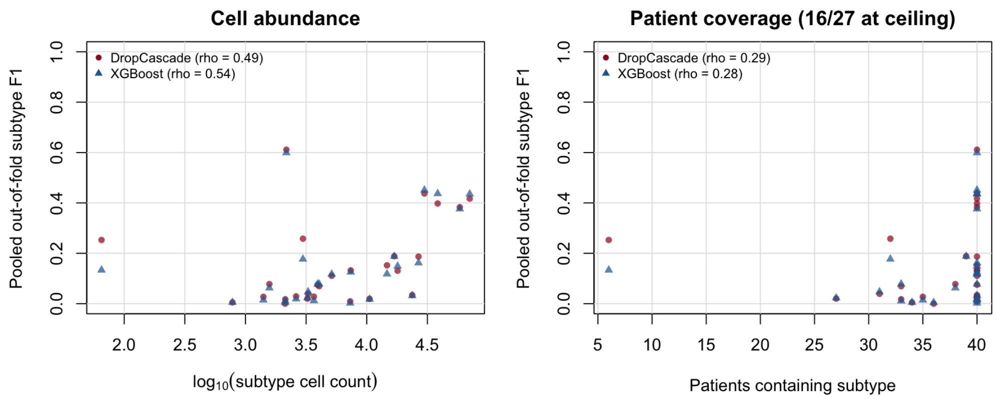

Figure 2: Per-subtype pooled out-of-fold F1 versus cell abundance (left) and patient coverage (right). Each marker is one of 27 subtypes; both models use the same ground-truth support values. Spearman correlations are marginal, and 16 of 27 subtypes occur in all 40 patients.

(a) Support–performance associations

Model	ρ_c	ρ_p	$\rho_{c\|p}$	$\rho_{p\|c}$
Soft-cascade Transformer	0.486	0.288	0.411	-0.051
Flat FT-style Transformer	0.505	0.255	0.464	-0.123
FT-style Transformer + HCE	0.513	0.318	0.427	-0.034
XGBoost	0.540	0.275	0.496	-0.130
Logistic Regression	0.435	0.107	0.488	-0.267
Linear SVM	0.339	-0.012	0.463	-0.335
Random Forest	0.661	0.418	0.564	-0.035
Reference Correlation	0.376	0.003	0.499	-0.354

(b) Granularity and error structure

Model	G	R_{sibling}	Cross-lineage
Soft-cascade Transformer	0.174	0.250	0.750
Flat FT-style Transformer	0.176	0.243	0.757
FT-style Transformer + HCE	0.183	0.257	0.743
XGBoost	0.179	0.246	0.754
Logistic Regression	0.167	0.295	0.705
Linear SVM	0.200	0.265	0.735
Random Forest	0.158	0.191	0.809
Reference Correlation	0.099	0.225	0.775

Table 12: Exploratory subtype diagnostics on cell-weighted pooled patient-disjoint predictions. ρ_c and ρ_p are marginal Spearman associations of subtype F1 with log cell count and patient coverage; partial correlations residualize ranked variables. $G = F_1^{\text{coarse}} - F_1^{\text{fine}}$. Among errors, R_{sibling} retains the true parent and Cross-lineage is its complement. These cohort-level summaries are descriptive, not causal.

Table 13 gives all subtype-level quantities and 95% percentile intervals from 2,000 patient-clustered bootstrap resamples. Counts and coverage use pooled ground truth, while F1 uses the same patient-disjoint OOF predictions as Table 2. Mean cells per patient averages only over patients in which the subtype is observed.

Table 13: Per-subtype support and pooled out-of-fold F1 with 95% percentile intervals from 2,000 patient-clustered bootstrap resamples. Cells/patient is the mean among patients in which that subtype is observed. Head and Tail denote the seven most and seven least abundant subtypes; all others are Mid.

Subtype	**Lineage**	**Cells**	**Patients**	**Cells/patient**	**Regime**	**Soft-cascade F1 [95% CI]**	**XGB F1 [95% CI]**
Erythrocyte	Other	65	6	10.8	Tail	0.253 [0.000, 0.395]	0.133 [0.000, 0.545]
Atypical Memory B	B	781	34	23.0	Tail	0.005 [0.000, 0.014]	0.005 [0.000, 0.014]
CD4 Monocyte-4	Myeloid	1,405	35	40.1	Tail	0.027 [0.006, 0.053]	0.014 [0.000, 0.033]
platelet	Other	1,574	38	41.4	Tail	0.078 [0.016, 0.125]	0.063 [0.015, 0.106]
Tcm	T	2,121	36	58.9	Tail	0.000 [0.000, 0.000]	0.004 [0.000, 0.016]
Memory B-2	B	2,130	33	64.5	Tail	0.017 [0.004, 0.033]	0.011 [0.000, 0.024]
Plasma	B	2,173	40	54.3	Tail	0.611 [0.522, 0.684]	0.599 [0.496, 0.685]
pDC	Myeloid	2,615	40	65.4	Mid	0.029 [0.010, 0.061]	0.019 [0.005, 0.040]
CD14 Monocyte-3	Myeloid	2,977	32	93.0	Mid	0.258 [0.170, 0.306]	0.177 [0.103, 0.211]
CD8 + CTL-3	T	3,244	27	120.1	Mid	0.020 [0.004, 0.035]	0.023 [0.008, 0.043]
GCB	B	3,283	31	105.9	Mid	0.039 [0.007, 0.071]	0.047 [0.001, 0.073]
Cycling T	T	3,667	40	91.7	Mid	0.028 [0.015, 0.043]	0.012 [0.003, 0.022]
CD14 Monocyte-2	Myeloid	3,931	40	98.3	Mid	0.075 [0.046, 0.102]	0.078 [0.056, 0.097]
CD8 + CTL-2	T	4,049	33	122.7	Mid	0.069 [0.017, 0.116]	0.078 [0.015, 0.113]
CD16 Monocyte-2	Myeloid	5,142	40	128.6	Mid	0.111 [0.064, 0.161]	0.117 [0.063, 0.164]
CD8 + CTL-1	T	7,303	40	182.6	Mid	0.009 [0.000, 0.024]	0.002 [0.001, 0.004]
cDC2	Myeloid	7,376	40	184.4	Mid	0.132 [0.099, 0.181]	0.125 [0.079, 0.167]
CD8 Tem-2	T	10,548	40	263.7	Mid	0.019 [0.005, 0.038]	0.017 [0.008, 0.030]
CD16 Monocyte-1	Myeloid	14,678	40	366.9	Mid	0.152 [0.111, 0.198]	0.118 [0.090, 0.150]
CD4 + Tcm-2	T	16,861	39	432.3	Mid	0.188 [0.111, 0.236]	0.188 [0.136, 0.220]
Naive T	T	17,960	40	449.0	Head	0.131 [0.067, 0.183]	0.149 [0.073, 0.199]
Memory B-1	B	23,722	40	593.0	Head	0.034 [0.020, 0.048]	0.032 [0.017, 0.047]
CD8 Tem-1	T	26,714	40	667.9	Head	0.187 [0.128, 0.250]	0.162 [0.107, 0.221]
CD14 Monocyte-1	Myeloid	29,871	40	746.8	Head	0.437 [0.378, 0.495]	0.451 [0.392, 0.507]
CD56 NK	NK	38,442	40	961.0	Head	0.398 [0.320, 0.436]	0.437 [0.373, 0.480]
CD16 NK	NK	58,507	40	1462.7	Head	0.383 [0.326, 0.434]	0.376 [0.317, 0.429]
CD4 + Tcm-1	T	70,653	40	1766.3	Head	0.417 [0.368, 0.460]	0.435 [0.393, 0.473]

For continuity with common long-tail reporting, Table 14 aggregates the seven least and seven most abundant subtypes. This summary is secondary to the per-class audit because individual subtypes can depart sharply from the abundance trend.

Model	**Tail Macro-F1**	**Head Macro-F1**
Soft-cascade Transformer	0.142	0.284
Flat FT-style Transformer	0.117	0.290
FT-style Transformer + HCE	0.125	0.288
XGBoost	0.119	0.292
Logistic Regression	0.114	0.197
Linear SVM	0.082	0.204
Random Forest	0.072	0.245
Reference Correlation	0.060	0.114

Table 14: Fine-subtype macro-F1 on the seven least and seven most abundant subtypes under cell-weighted pooled patient-disjoint evaluation. This aggregate is reported as a descriptive complement to the complete per-subtype audit in Table 13.

Cross-lineage error relative to an error-conditioned null. Table 15 uses the wrong-conditioned null as the primary reference: after excluding the true subtype, it renormalizes the predicted-subtype marginal so every randomized outcome remains an error. For the four main models, observed cross-lineage rates are 0.743–0.757 versus wrong-conditioned expectations of 0.741–0.745. The corresponding differences are only +0.002 to +0.013, and all four patient-bootstrap 95% intervals include zero. Thus the raw approximately 75% rate is largely explained by the available wrong-class distribution and does not independently demonstrate either hierarchy preservation or hierarchy failure. The unconditional permutation, which can create chance exact matches and therefore uses a different sample space, is retained only as a sensitivity analysis.

Model	Errors	Observed	Wrong-cond. null	Obs.−WC null [95% CI]	Uncond. null (sens.)
Soft-cascade Transformer	251,354	0.750	0.744	+0.005 [-0.025, +0.028]	0.675
Flat FT-style Transformer	247,798	0.757	0.745	+0.011 [-0.020, +0.035]	0.675
FT-style Transformer + HCE	250,944	0.743	0.741	+0.002 [-0.028, +0.023]	0.671
XGBoost	244,990	0.754	0.741	+0.013 [-0.020, +0.038]	0.672
Logistic Regression	301,566	0.705	0.752	-0.047 [-0.061, -0.036]	0.720
Linear SVM	296,904	0.735	0.774	-0.039 [-0.053, -0.028]	0.738
Random Forest	246,420	0.809	0.749	+0.059 [+0.021, +0.083]	0.685
Reference Correlation	328,596	0.775	0.785	-0.009 [-0.022, +0.003]	0.758

Table 15: Cross-lineage error relative to an error-conditioned marginal null. The primary wrong-conditioned (WC) null draws from each model's predicted-subtype marginal after excluding the true subtype and renormalizing, so every randomized outcome remains wrong. Observed−WC-null confidence intervals recompute both terms in 2,000 patient-clustered bootstrap resamples. The unconditional permutation preserves the realized true- and predicted-subtype marginals but can create chance exact matches; it is reported only as a sensitivity analysis.

C.9 Protocol Sensitivity: Patient-Level versus Cell-Level Splitting

Table 16 compares the primary patient-disjoint protocol with a leakage-prone cell-level split control. It is placed in the appendix because the central result uses patient-disjoint evaluation throughout; the control quantifies how much the easier protocol would inflate both absolute performance and the apparent advantage of a model family.

Metric definitions. Let $C^{(p)} \in \mathbb{R}^{K\times K}$ be the confusion matrix for held-out subject p and $n_p = \sum_{ij} C^{(p)}_{ij}$. We distinguish

$$C^{\mathrm{CW}} = \sum_p C^{(p)}, \qquad F_1^{\mathrm{CW}} = \mathrm{MacroF1}(C^{\mathrm{CW}}), \tag{11}$$

$$C^{\mathrm{SB}} = \frac{1}{P}\sum_p \frac{C^{(p)}}{n_p}, \qquad F_1^{\mathrm{SB}} = \mathrm{MacroF1}(C^{\mathrm{SB}}), \tag{12}$$

$$F_1^{\mathrm{within}} = \frac{1}{P}\sum_p \mathrm{MacroF1}(C^{(p)}). \tag{13}$$

We call these *cell-weighted pooled Macro-F1*, *subject-balanced pooled Macro-F1*, and *mean within-subject Macro-F1*. Subject balancing normalizes each subject confusion matrix to unit mass before pooling; it is neither equal-cell subsampling nor averaging per-subject Macro-F1. Released files retain the legacy column name `patient_balanced_macro_f1` for F_1^{SB}, but no different patient-balanced statistic is used.

Task	Model	Patient-disjoint	Cell split	Inflation
Coarse lineage	XGBoost	0.322	0.372	+0.050
	Soft-cascade Transformer	0.326	0.417	+0.090
Fine subtype	XGBoost	0.143	0.212	+0.069
	Soft-cascade Transformer	0.152	0.260	+0.108

Table 16: Macro-F1 under the primary patient-disjoint protocol versus a leakage-prone cell-level split control. Cell-level splitting inflates performance for both methods, and does so more strongly for Soft-cascade Transformer than for XGBoost. This demonstrates that evaluation protocol can change both absolute performance and the apparent relative advantage of model families, which is why APT-Bench treats patient-disjoint evaluation as a requirement rather than a reporting option.

Equal-cell evaluation sensitivity. Patient-disjoint evaluation prevents leakage but does not make a cell-weighted pooled point estimate subject-balanced; patient cell counts range from 1,929 to 22,627. Table 17 therefore recomputes both hierarchy endpoints after sampling 1,900 cells without replacement from every patient for each of 50 seeds. Coarse macro-F1 changes by at most $+0.005$, while fine macro-F1 changes by -0.007 to -0.009; the qualitative model ordering is unchanged. The main conclusions are therefore not driven by a few high-cell-count patients, although the headline pooled metric remains explicitly cell-weighted.

Task	**Model**	**CW-pooled M-F1**	**Equal-cell M-F1 [95% interval]**	**Δ**
Coarse lineage	Soft-cascade Transformer	0.326	0.331 [0.325, 0.337]	+0.005
	Flat FT-style Transformer	0.323	0.326 [0.322, 0.330]	+0.003
	FT-style Transformer + HCE	0.333	0.334 [0.328, 0.340]	+0.000
	XGBoost	0.322	0.324 [0.320, 0.329]	+0.002
Fine subtype	Soft-cascade Transformer	0.152	0.144 [0.138, 0.153]	-0.008
	Flat FT-style Transformer	0.147	0.138 [0.131, 0.147]	-0.008
	FT-style Transformer + HCE	0.151	0.142 [0.137, 0.151]	-0.009
	XGBoost	0.143	0.136 [0.131, 0.144]	-0.007

Table 17: Equal-cell evaluation sensitivity. For each of 50 seeds, 1,900 cells are sampled without replacement from every subject, and cell-weighted pooled Macro-F1 is recomputed on the resulting 76,000 cells. All models use identical sampled row indices. The interval is the empirical 2.5th–97.5th percentile across seeds; Δ is the equal-cell mean minus the original cell-weighted pooled metric. This resampling diagnostic is not the subject-balanced pooled metric, and it does not retrain or reselect models.

C.10 Fair Fixed-Total and Matched-Scale Analysis

The primary scaling design uses $P \in \{8, 16, 32\}$ and $C \in \{100, 200, 400, 800, 1600\}$, with 20 nested subset seeds for every outer fold. Because every retained patient has at least 1,929 cells, all finite caps are feasible without replacement. Disease-stratified patient orderings are nested across P, uniform cell permutations are nested across C, no subtype label enters sampling, and preprocessing is refit on each sampled training set. The fixed-total diagonals satisfy $PC \in \{3200, 6400, 12800\}$ exactly.

Model	Task	Total cells	$P = 8$	$P = 16$	$P = 32$
LR	Coarse	3,200	0.270 [0.259, 0.280]	0.277 [0.270, 0.285]	0.282 [0.279, 0.286]
LR	Coarse	6,400	0.277 [0.266, 0.290]	0.286 [0.278, 0.293]	0.292 [0.290, 0.295]
LR	Coarse	12,800	0.281 [0.270, 0.294]	0.291 [0.284, 0.300]	0.298 [0.296, 0.301]
LR	Fine	3,200	0.078 [0.073, 0.084]	0.082 [0.079, 0.087]	0.085 [0.083, 0.089]
LR	Fine	6,400	0.080 [0.077, 0.087]	0.085 [0.081, 0.089]	0.089 [0.087, 0.091]
LR	Fine	12,800	0.081 [0.076, 0.087]	0.087 [0.083, 0.090]	0.091 [0.089, 0.094]
XGBoost	Coarse	3,200	0.262 [0.250, 0.276]	0.272 [0.260, 0.284]	0.279 [0.274, 0.284]
XGBoost	Coarse	6,400	0.269 [0.258, 0.284]	0.279 [0.267, 0.289]	0.288 [0.285, 0.292]
XGBoost	Coarse	12,800	0.276 [0.267, 0.290]	0.289 [0.277, 0.298]	0.298 [0.296, 0.300]
XGBoost	Fine	3,200	0.082 [0.078, 0.088]	0.087 [0.081, 0.092]	0.089 [0.086, 0.092]
XGBoost	Fine	6,400	0.086 [0.081, 0.094]	0.091 [0.086, 0.097]	0.095 [0.090, 0.098]
XGBoost	Fine	12,800	0.092 [0.087, 0.098]	0.097 [0.092, 0.102]	0.098 [0.096, 0.101]

Table 18: Fixed-total training-cell scaling. Each row holds the total number of sampled training cells constant while reallocating them across independent patients. Entries are subject-balanced pooled, patient-disjoint OOF Macro-F1 means with empirical 95% intervals over 20 matched subset seeds. For each score, every held-out subject's confusion matrix is normalized to unit mass before pooling; this is distinct from both cell-weighted pooling and mean within-subject Macro-F1. All outer-test cells are evaluated.

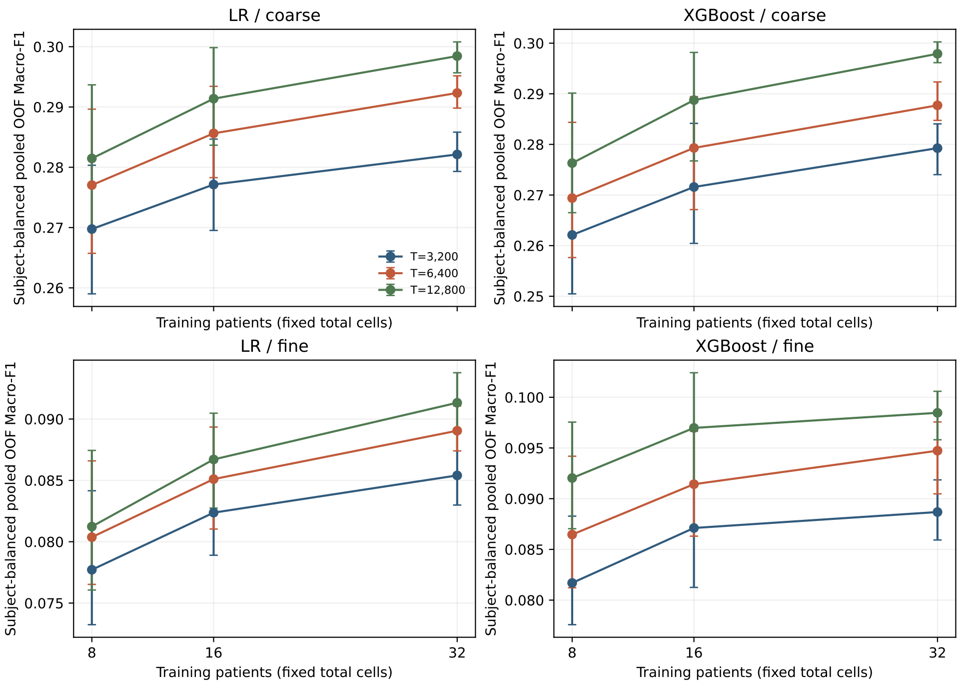

Figure 3: APT-only fixed-total response curves underlying the unmatched cross-cohort diagnostic in Table 19. Points are pooled OOF macro-F1 means over 20 matched subset seeds; bars are empirical 95% seed intervals. Patient-balanced and patient-bootstrap results are provided in the machine-readable artifacts.

Table 18 reports variability across subset seeds. Separately, paired patient-clustered bootstraps over the same 40 pooled OOF patients compare $P = 8$ with $P = 32$ after averaging seed-level sufficient statistics. All 12 intervals exclude zero; their lower endpoints range from 0.0002 to 0.0084 and upper endpoints from 0.0090 to 0.0314. These intervals quantify uncertainty over patients, whereas the empirical seed intervals quantify sensitivity to which development patients and cells are sampled.

Matched doubling effects average patient versus cell gains of 0.011/0.005 (LR coarse), 0.007/0.003 (LR fine), 0.016/0.008 (XGBoost coarse), and 0.009/0.006 (XGBoost fine). In the descriptive fold-fixed-effect response surface, the corresponding per-doubling β_P/β_C coefficients are 0.017/0.008, 0.008/0.004, 0.016/0.007, and 0.009/0.006. Complete pairwise effects, patient intervals, seed intervals, and response-surface uncertainty are stored in the machine-readable scaling artifacts. The coefficients are associations over this grid, not causal or universal scaling laws.

C.11 External COMBAT and OneK1K Scaling Replications

We downloaded the author-released COMBAT CITE-seq object (COVID-19 Multi-omics Blood ATlas (COMBAT) Consortium, 2022) (836,148 cells and 124 donor identifiers; SHA-256 `f628a15f25b9ca2f7cdefeab271fd9d007c8d5e47eb80c4b807d5f65e86ff53d`). We removed cells without a resolved major or fine immune annotation and cells flagged as GEX-region doublets. Ten donors were frozen as a calibration-only set and excluded from every model fit and test fold. Among the remaining donors, requiring at least 1,600 eligible cells leaves 105 donors and 700,706 cells. We construct five disjoint folds of 21 test donors, stratified by the released sample-source field, and freeze these folds across all models, budgets, seeds, and modalities.

The RNA replication uses 293 protein-coding genes with highest variance on the 10 calibration donors after excluding mitochondrial and ribosomal genes. This matches the APT input dimension without selecting features on any evaluation donor. The ADT sensitivity uses all 192 released antibody features. Both inputs use the released five-class major annotation and 39-class fine annotation. They share the exact P, C, and fixed-total grids, 20 nested source-stratified donor/cell seeds, training-only standardization, LR configuration, and paired donor-bootstrap procedure used in APT-Bench. The RNA analysis additionally uses the frozen 400-tree XGBoost baseline. Every fit is evaluated on all cells of the 21 untouched test donors; the ADT analysis is an LR-only modality sensitivity rather than an additional model-comparison family. Dataset hashes, folds, features, sampling manifests, and per-donor sufficient statistics are included in the release artifacts.

OneK1K provides a second independent scRNA-seq replication in a much larger healthy PBMC cohort (Yazar et al., 2022). We use the current author-deposited CELLxGENE object (1,248,980 cells and 981 donor identifiers; SHA-256 `14c5e9e70007a5e7dc047f7b29405773305e1a7b9791d0789af70649cf6f46ec`), remove released doublet calls, and reserve one complete multiplexing pool for calibration. Requiring at least 800 eligible cells leaves 925 evaluation donors and 1,199,468 cells. Five folds are disjoint in both donor and multiplexing pool. We select 293 protein-coding, non-mitochondrial, non-ribosomal genes by variance on ten calibration-pool donors only, and use the released 30-class Azimuth-derived fine labels with a deterministic five-lineage mapping.

OneK1K uses the same models, 20 nested subset seeds, training-only standardization, untouched-test-donor evaluation, subject-balanced pooled metric, and 2,000 paired donor bootstraps. Its grid is $P \in \{8, 16, 32\}$ and $C \in \{100, 200, 400, 800\}$, yielding the two common exact budgets $T \in \{3200, 6400\}$. We prespecified the lower depth ceiling because only 116 of the 981 raw donors have at least 1,600 eligible cells; enforcing the full APT grid would discard most donors and condition the replication on high cell recovery. OneK1K tests scaling portability in a healthy cohort, not disease generalization or absolute accuracy across different feature spaces and label-generation procedures.

Dataset	Modality	Model/task	$T = 3{,}200$	$T = 6{,}400$	$T = 12{,}800$
APT	Aptamer	LR/Coarse	+0.012 [+0.005, +0.021]	+0.015 [+0.007, +0.024]	+0.017 [+0.009, +0.026]
APT	Aptamer	LR/Fine	+0.008 [+0.004, +0.011]	+0.009 [+0.005, +0.012]	+0.010 [+0.005, +0.013]
APT	Aptamer	XGBoost/Coarse	+0.017 [+0.008, +0.027]	+0.018 [+0.007, +0.030]	+0.022 [+0.009, +0.035]
APT	Aptamer	XGBoost/Fine	+0.007 [+0.002, +0.011]	+0.008 [+0.002, +0.013]	+0.006 [+0.000, +0.011]
COMBAT	RNA	LR/Coarse	+0.003 [+0.002, +0.004]	+0.002 [+0.001, +0.003]	+0.001 [+0.001, +0.002]
COMBAT	RNA	LR/Fine	+0.012 [+0.009, +0.013]	+0.012 [+0.010, +0.012]	+0.007 [+0.005, +0.009]
COMBAT	RNA	XGBoost/Coarse	+0.006 [+0.005, +0.007]	+0.005 [+0.004, +0.006]	+0.004 [+0.003, +0.005]
COMBAT	RNA	XGBoost/Fine	+0.016 [+0.013, +0.017]	+0.018 [+0.016, +0.019]	+0.017 [+0.014, +0.018]
COMBAT	ADT	LR/Coarse	+0.006 [+0.004, +0.008]	+0.009 [+0.006, +0.012]	+0.012 [+0.010, +0.016]
COMBAT	ADT	LR/Fine	+0.013 [+0.010, +0.014]	+0.011 [+0.010, +0.013]	+0.011 [+0.009, +0.013]
OneK1K	RNA	LR/Coarse	+0.004 [+0.004, +0.005]	+0.005 [+0.005, +0.006]	–
OneK1K	RNA	LR/Fine	+0.007 [+0.007, +0.009]	+0.008 [+0.007, +0.009]	–
OneK1K	RNA	XGBoost/Coarse	-0.003 [-0.004, -0.002]	+0.006 [+0.005, +0.006]	–
OneK1K	RNA	XGBoost/Fine	+0.006 [+0.005, +0.007]	+0.009 [+0.007, +0.009]	–

Table 19: Unmatched diagnostic: change in subject-balanced pooled, subject-disjoint OOF Macro-F1 when the same total training-cell budget is reallocated from 8 to 32 independent subjects without freezing the marginal training label distribution. Each held-out subject confusion matrix is normalized to unit mass before pooling. Brackets are 95% paired subject-bootstrap intervals over the fixed outer-test subjects. Positive values favor broader subject coverage but can also reflect changes in observed classes, per-class counts, and class proportions. COMBAT ADT is a prespecified modality sensitivity using LR only. OneK1K uses the two common exact budgets supported by at least 800 cells from nearly all donors.

C.12 Label-Distribution Audit and Matched Control

The label-agnostic grid samples cells uniformly within subjects and therefore allows the full marginal label distribution to vary. Machine-readable audit tables report the observed class count, unseen-at-training class count, cell count, and donor count for every class at every fold, seed, and scaling point. The main text reports the common fixed-total diagonals, where fine-class coverage increases slightly with P despite fixed T; this coverage statistic is only one observable component of the broader label-distribution effect.

The class-matched control fixes the marginal label distribution. The $P = 8$ subset defines the observed labels and a proportional integer quota T_k with a one-cell floor; capacity violations are deterministically redistributed. The same quota vector is then used at $P = 8, 16, 32$. Within each class, a capacity-constrained round-robin sampler distributes cells across selected subjects before assigning a second cell from the same subject, so increasing P changes realized donor coverage rather than class counts. Sampling is without replacement, standardization is refit on each training subset, and all immutable outer-test cells are retained. All three runs complete 1,200/1,200 jobs and pass checks for exact total cells, identical paired class counts, non-overlapping test subjects, and contribution by every nominal training subject. These primary Logistic Regression runs use 20 matched subset seeds at both $T = 3{,}200$ and $T = 6{,}400$. To test whether the corrected direction is specific to a linear classifier, we repeat the identical control with the frozen XGBoost configuration at $T = 6{,}400$ using 10 matched seeds. Each dataset completes 300/300 additional jobs and passes the same checks; final production fits use the CPU histogram backend throughout to avoid mixing computational backends. It does not separately identify changes in disease–subtype composition, within-class allocation across subjects, or within-class subject heterogeneity; these remain bundled with the subject-breadth intervention.

Dataset	Model	Task	T	Classes (mean/total)	Donors/class (P: 8→32)	Δ SB-pooled M-F1	$\Pr(\Delta > 0)$
APT	LR	Coarse	3,200	5.0/5	7.9→28.5	+0.0062 [-0.0080, +0.0221]	0.778
APT	LR	Coarse	6,400	5.0/5	7.9→30.5	+0.0078 [-0.0070, +0.0265]	0.836
APT	LR	Fine	3,200	26.8/27	7.1→25.0	-0.0024 [-0.0077, +0.0029]	0.163
APT	LR	Fine	6,400	26.8/27	7.2→27.4	-0.0050 [-0.0105, +0.0009]	0.043
APT	XGBoost	Coarse	6,400	5.0/5	7.9→30.5	+0.0083 [-0.0130, +0.0265]	0.817
APT	XGBoost	Fine	6,400	26.8/27	7.2→27.3	-0.0058 [-0.0143, +0.0027]	0.110
COMBAT RNA	LR	Coarse	3,200	5.0/5	8.0→29.0	+0.0011 [-0.0036, +0.0084]	0.611
COMBAT RNA	LR	Coarse	6,400	5.0/5	8.0→31.3	+0.0000 [-0.0055, +0.0045]	0.547
COMBAT RNA	LR	Fine	3,200	39.0/39	7.2→22.5	-0.0007 [-0.0088, +0.0092]	0.439
COMBAT RNA	LR	Fine	6,400	39.0/39	7.3→26.6	-0.0028 [-0.0114, +0.0043]	0.272
COMBAT RNA	XGBoost	Coarse	6,400	5.0/5	8.0→31.2	+0.0011 [-0.0054, +0.0088]	0.522
COMBAT RNA	XGBoost	Fine	6,400	39.0/39	7.3→26.6	-0.0027 [-0.0096, +0.0066]	0.264
OneK1K RNA	LR	Coarse	3,200	5.0/5	7.8→27.7	+0.0025 [-0.0079, +0.0120]	0.669
OneK1K RNA	LR	Coarse	6,400	5.0/5	7.8→29.6	+0.0033 [-0.0059, +0.0128]	0.728
OneK1K RNA	LR	Fine	3,200	29.0/30	6.2→19.8	+0.0019 [-0.0080, +0.0105]	0.700
OneK1K RNA	LR	Fine	6,400	29.0/30	6.4→22.1	+0.0013 [-0.0058, +0.0089]	0.598
OneK1K RNA	XGBoost	Coarse	6,400	5.0/5	7.8→29.5	-0.0011 [-0.0061, +0.0139]	0.209
OneK1K RNA	XGBoost	Fine	6,400	28.9/30	6.4→22.0	-0.0006 [-0.0117, +0.0098]	0.549

Table 20: Class-matched fixed-total control. The outcome is subject-balanced pooled Macro-F1, computed by normalizing each held-out subject confusion matrix to unit mass before pooling. LR uses 20 seeds at both totals; the matched XGBoost robustness check uses 10 seeds at $T = 6{,}400$. Within each outer-fold/seed/task pair, the $P = 8$ subset freezes the observed classes and exact per-class cell quotas; the same quotas are used at $P = 16$ and $P = 32$. Brackets are joint 95% intervals obtained by resampling a training-subset seed and then paired outer-test subjects; the final column is the empirical fraction of joint replicates with a positive effect. Seed-only and subject-only intervals are released separately. Donors/class reports the mean number of subjects contributing cells to an observed class.

The primary interval propagates both uncertainty sources. In each of 2,000 replicates, we sample one completed training-subset seed uniformly, resample outer-test subjects with replacement, use the same subject draw for $P = 8$ and $P = 32$, and recompute the paired difference. All 18 joint 95% intervals include zero, and the table reports the corresponding $\Pr(\Delta > 0)$. For decomposition, empirical seed-only intervals include zero in 16 of 18 rows; the exceptions are the two negative APT fine results at $T = 6{,}400$. Subject-only intervals condition on the seed-averaged prediction matrix and can exclude zero even when the joint interval does not. We therefore treat the joint interval as primary and interpret the conditional intervals only as diagnostics of the two uncertainty sources.

C.13 Extended External Subject Frontier

The common $P \leq 32$ grid is imposed by the 40-patient APT cohort, not by the capacity of the external datasets. We therefore extend the class-matched LR control to $P = 64$ for COMBAT and to $P \in \{64, 128\}$ for OneK1K at $T \in \{3{,}200, 6{,}400\}$. The original $P = 8$ subset freezes the class set and integer per-class quotas for every fold, seed, task, and total; only the number of selected subjects allowed to contribute those cells changes. Each new point uses 10 nested subset seeds, all immutable outer-test subjects, and the same joint resampling of training-subset seed and held-out subjects used in the primary matched control. This extension is an external-cohort allocation sensitivity rather than a claim that its largest tested P is globally optimal.

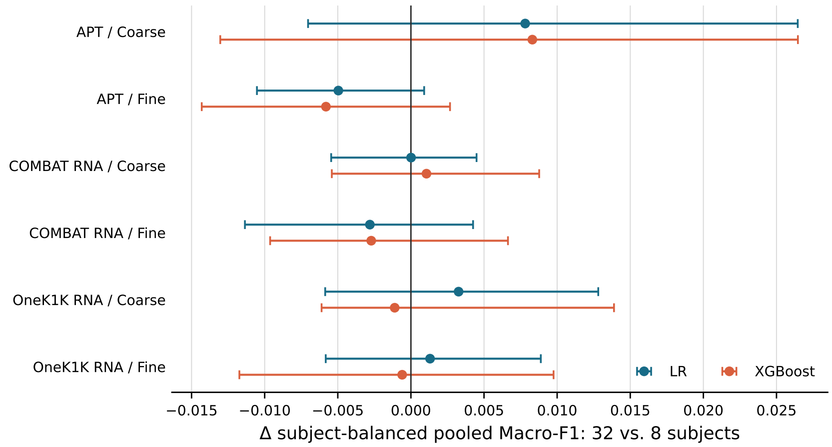

Figure 4: Joint uncertainty for the completed class-matched protocol at $T = 6{,}400$. Points are mean changes in subject-balanced pooled OOF macro-F1 from 8 to 32 subjects. The 95% joint intervals select a training-subset seed and bootstrap paired held-out subjects; all 12 span zero. Unlike the seed-only bars in Figure 1, these intervals include test-subject variability. Zero inclusion does not establish equivalence.

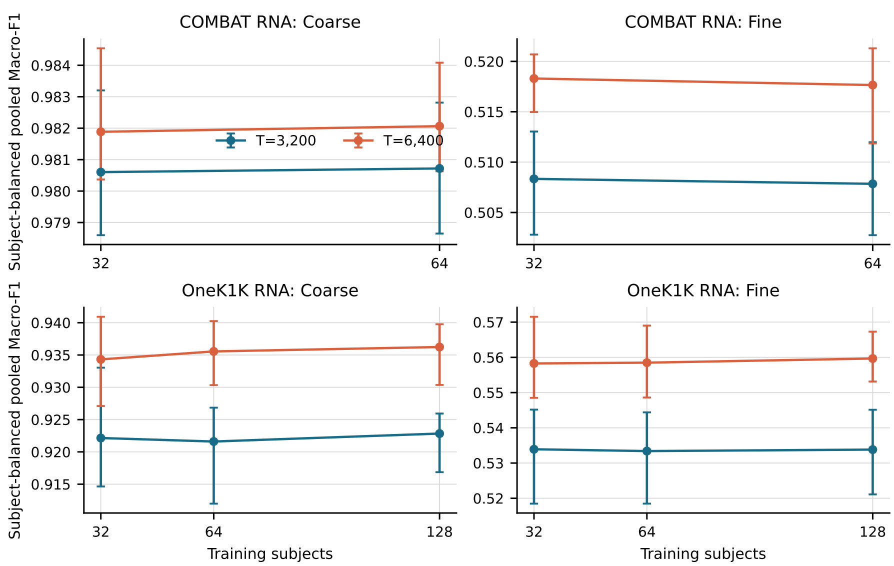

Figure 5: Class-matched LR frontier beyond 32 training subjects. Points are mean subject-balanced pooled OOF macro-F1 over 10 matched training-subset seeds; bars are empirical 95% seed intervals and therefore visualize subset sensitivity, not the primary joint uncertainty. The paired joint effect intervals are reported in Table 21.

Dataset	Task	T	Subjects	Δ SB-pooled M-F1	$\Pr(\Delta > 0)$
COMBAT RNA	Coarse	3,200	32→64	+0.0001 [-0.0020, +0.0025]	0.535
COMBAT RNA	Coarse	6,400	32→64	+0.0002 [-0.0014, +0.0016]	0.618
COMBAT RNA	Fine	3,200	32→64	-0.0005 [-0.0060, +0.0026]	0.498
COMBAT RNA	Fine	6,400	32→64	-0.0006 [-0.0060, +0.0040]	0.450
OneK1K RNA	Coarse	3,200	32→64	-0.0005 [-0.0111, +0.0052]	0.553
OneK1K RNA	Coarse	6,400	32→64	+0.0012 [-0.0046, +0.0046]	0.766
OneK1K RNA	Fine	3,200	32→64	-0.0005 [-0.0060, +0.0053]	0.418
OneK1K RNA	Fine	6,400	32→64	+0.0002 [-0.0096, +0.0063]	0.595
OneK1K RNA	Coarse	3,200	64→128	+0.0012 [-0.0023, +0.0068]	0.642
OneK1K RNA	Coarse	6,400	64→128	+0.0007 [-0.0042, +0.0043]	0.657
OneK1K RNA	Fine	3,200	64→128	+0.0004 [-0.0039, +0.0050]	0.542
OneK1K RNA	Fine	6,400	64→128	+0.0012 [-0.0066, +0.0068]	0.617

Table 21: Extended class-matched subject frontier for Logistic Regression. Each comparison holds the total training-cell budget, observed class set, and exact per-class cell quotas fixed while increasing the number of contributing training subjects. Values are paired mean changes; brackets are joint 95% intervals from 2,000 replicates that sample a matched training-subset seed and then bootstrap held-out subjects. The final column is the empirical joint probability of a positive change.

All 12 adjacent frontier effects are small: their means range from -0.00065 to $+0.00123$, and every joint interval includes zero. The four direct OneK1K $32 \to 128$ contrasts are likewise unresolved. This does not by itself establish a plateau or exclude practically meaningful changes. The explicit margin sensitivity below assesses interval width separately from zero inclusion. Because the intervention remains tied to these tasks, budgets, features, and LR configuration, it does not locate a universal saturation point or optimal allocation.

C.14 Practical-Equivalence Sensitivity

Failure to detect a difference is not evidence of equivalence (Lakens, 2017). After inspecting the existing results, we add an *exploratory* smallest-effect sensitivity bound $\epsilon = 0.01$ absolute subject-balanced pooled Macro-F1 (one percentage point), with 0.005 and 0.02 as sensitivity values. This is a reporting tolerance, not a validated clinical threshold, and was not prespecified for the completed experiments. It must not be conflated with the separate compact-panel tolerance $\delta = 0.02$.

For each released joint 95% interval $[L, U]$, we report zero inclusion ($L \leq 0 \leq U$) and strict containment ($-\epsilon < L$ and $U < \epsilon$) separately. Containment is evidence of practical equivalence only relative to that margin, interval construction, and estimand; zero inclusion alone is not. Intervals with $L \leq -\epsilon$ and $U \geq \epsilon$ lack precision to exclude meaningful effects in either direction. Other non-contained intervals can still leave a meaningful one-sided effect unresolved. These are not mutually exclusive categories, and interval width alone does not establish formal statistical power; we therefore use "insufficient precision" rather than claiming a quantified underpowered design.

At $T = 6{,}400$, only 4 of 12 main comparison intervals meet the ± 0.01 containment criterion; none of the four APT intervals does. The APT coarse XGBoost interval spans both meaningful directions. Among adjacent external LR frontier comparisons, 11 of 12 intervals meet this criterion; the exception is OneK1K coarse $32 \to 64$ at $T = 3{,}200$. With the tighter ± 0.005 margin, no main interval and only 4 of 12 adjacent frontier intervals are contained. Conclusions about practical equivalence are thus explicitly margin-dependent, not a blanket finding that subjects have no value.

We do not report formal TOST p-values. Conventional level-0.05 TOST is linked to a corresponding 90% confidence interval, whereas this analysis retains the released joint 95% intervals. Moreover, those intervals sample one completed training-subset seed and then test subjects, combining between-fit variability with test uncertainty rather than estimating only a seed-averaged mean effect. Containment is therefore an exploratory joint-interval screen, not a calibrated confirmatory equivalence test. Results are pointwise, without a simultaneous equivalence guarantee across rows, and do not establish a continuous plateau between or beyond the tested budgets. This audit concerns the completed legacy controls, not the ongoing A–D GPU rerun.

Comparison	Dataset	Task	n	$\pm.005$	$\pm.010$	$\pm.020$	Both signs*
8–32	APT	coarse	2	0	0	0	1
8–32	APT	fine	2	0	0	2	0
8–32	COMBAT	coarse	2	0	2	2	0
8–32	COMBAT	fine	2	0	1	2	0
8–32	OneK1K	coarse	2	0	0	2	0
8–32	OneK1K	fine	2	0	1	2	0
Extended LR	COMBAT	coarse	2	2	2	2	0
Extended LR	COMBAT	fine	2	0	2	2	0
Extended LR	OneK1K	coarse	4	2	3	4	0
Extended LR	OneK1K	fine	4	0	4	4	0

Table 22: Post-hoc practical-equivalence sensitivity: number of released joint 95% intervals strictly inside each stated margin. The 8–32 rows use $T = 6{,}400$ and both LR and XGBoost; extended rows use LR at both totals for adjacent subject-budget comparisons. *Intervals reaching both -0.01 and $+0.01$, indicating insufficient precision to exclude meaningful effects in either direction. Counts are pointwise diagnostics, not formal TOST decisions or evidence that an entire scaling frontier is equivalent.

Model	Task	$C = 100$	$C = 500$	$C = 2000$	$C = \text{all}$
Logistic Regression, $P = 8$	Coarse	0.255 [0.247, 0.265]	0.274 [0.263, 0.284]	0.283 [0.269, 0.297]	0.282 [0.269, 0.293]
Logistic Regression, $P = 16$	Coarse	0.266 [0.261, 0.270]	0.290 [0.284, 0.297]	0.288 [0.282, 0.295]	0.290 [0.280, 0.300]
Logistic Regression, $P = 24$	Coarse	0.276 [0.269, 0.280]	0.296 [0.288, 0.304]	0.288 [0.282, 0.295]	0.296 [0.289, 0.302]
Logistic Regression, $P = 32$	Coarse	0.283 [0.278, 0.288]	0.299 [0.296, 0.302]	0.287 [0.284, 0.288]	0.299 [0.284, 0.315]
Logistic Regression, $P = 8$	Fine	0.074 [0.068, 0.081]	0.084 [0.079, 0.088]	0.087 [0.082, 0.092]	0.101 [0.091, 0.109]
Logistic Regression, $P = 16$	Fine	0.081 [0.078, 0.085]	0.090 [0.088, 0.094]	0.095 [0.090, 0.099]	0.116 [0.110, 0.125]
Logistic Regression, $P = 24$	Fine	0.086 [0.083, 0.089]	0.093 [0.089, 0.097]	0.100 [0.095, 0.105]	0.125 [0.118, 0.131]
Logistic Regression, $P = 32$	Fine	0.089 [0.086, 0.092]	0.095 [0.093, 0.097]	0.105 [0.102, 0.108]	0.132 [0.112, 0.146]
XGBoost, $P = 8$	Coarse	0.246 [0.234, 0.263]	0.262 [0.250, 0.278]	0.277 [0.262, 0.293]	0.300 [0.283, 0.321]
XGBoost, $P = 16$	Coarse	0.260 [0.252, 0.272]	0.278 [0.268, 0.288]	0.295 [0.283, 0.306]	0.313 [0.298, 0.330]
XGBoost, $P = 24$	Coarse	0.270 [0.264, 0.276]	0.289 [0.284, 0.295]	0.305 [0.297, 0.312]	0.319 [0.311, 0.328]
XGBoost, $P = 32$	Coarse	0.275 [0.270, 0.280]	0.296 [0.293, 0.298]	0.310 [0.308, 0.312]	0.322 [0.300, 0.345]
XGBoost, $P = 8$	Fine	0.073 [0.068, 0.079]	0.085 [0.080, 0.091]	0.096 [0.090, 0.103]	0.118 [0.110, 0.126]
XGBoost, $P = 16$	Fine	0.082 [0.077, 0.085]	0.096 [0.092, 0.101]	0.110 [0.103, 0.120]	0.132 [0.125, 0.140]
XGBoost, $P = 24$	Fine	0.087 [0.081, 0.092]	0.100 [0.095, 0.104]	0.115 [0.110, 0.121]	0.138 [0.132, 0.144]
XGBoost, $P = 32$	Fine	0.090 [0.087, 0.093]	0.102 [0.100, 0.105]	0.118 [0.117, 0.121]	0.143 [0.127, 0.159]

Table 23: Patient–cell training scaling. Entries are pooled OOF Macro-F1 means with empirical 2.5th–97.5th seed intervals; the deterministic $P = 32, C = \text{all}$ endpoint instead uses a patient-clustered bootstrap interval. Seed intervals are not patient confidence intervals.

C.15 Secondary Descriptive Patient–Cell Scaling Surface

This historical 4×4 surface fixes eight outer-test patients and varies only the training set. For each outer fold and each of 20 seeds, we generate one approximately proportional disease-stratified ordering of the 32 development patients, giving nested sets $S_8 \subset S_{16} \subset S_{24} \subset S_{32}$. Each selected patient receives one uniform random cell ordering, giving nested prefixes of 100, 500, and 2,000 cells and the complete available set. Patients with fewer than a requested cap contribute all available cells. No cell is selected using its subtype, and every model is evaluated on all cells from the untouched test patients. The full-data endpoint is fitted once per fold rather than duplicated across seeds. All 6,020 model–task jobs completed, and the deterministic endpoint exactly reproduces the frozen pooled OOF baselines within absolute tolerance 5×10^{-4}. Because $P{=}8 \to 32$ is a fourfold change whereas $C{=}100 \to \text{all}$ is a larger and patient-dependent change, this surface is descriptive only. All direct comparisons between the two axes use the fixed-total and matched-doubling analysis in the main paper.

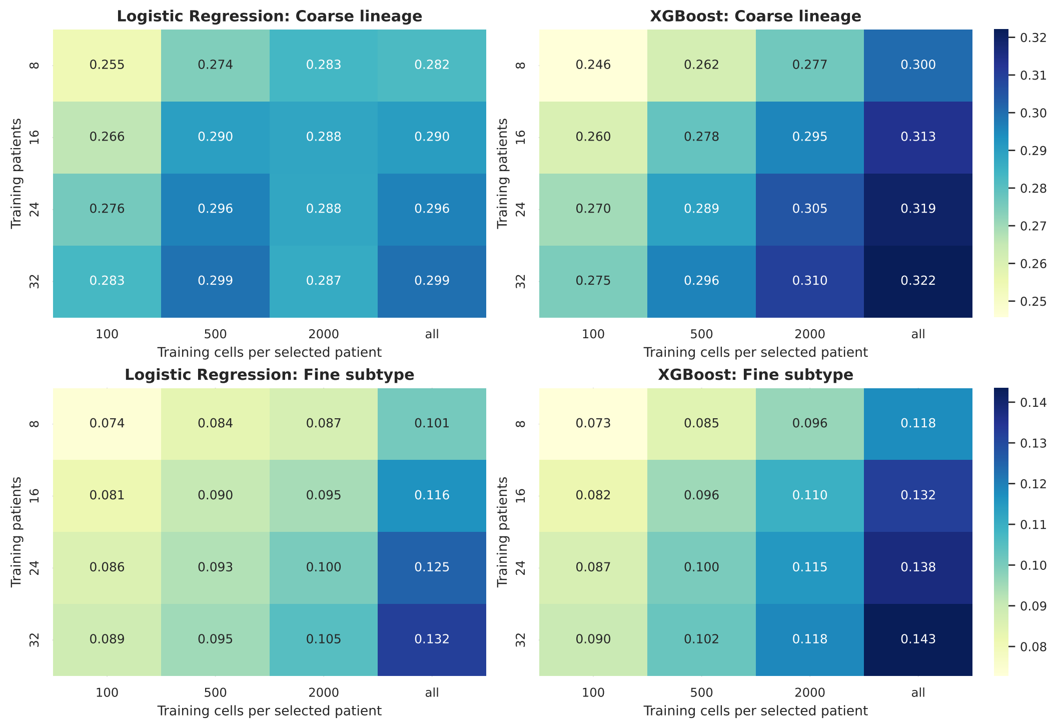

Figure 6: Cell-weighted pooled patient-disjoint Macro-F1 over the complete training-patient and per-patient training-cell grid. Values average 20 nested subset seeds except for the deterministic full-data endpoint. Color scales are shared across models within each task.

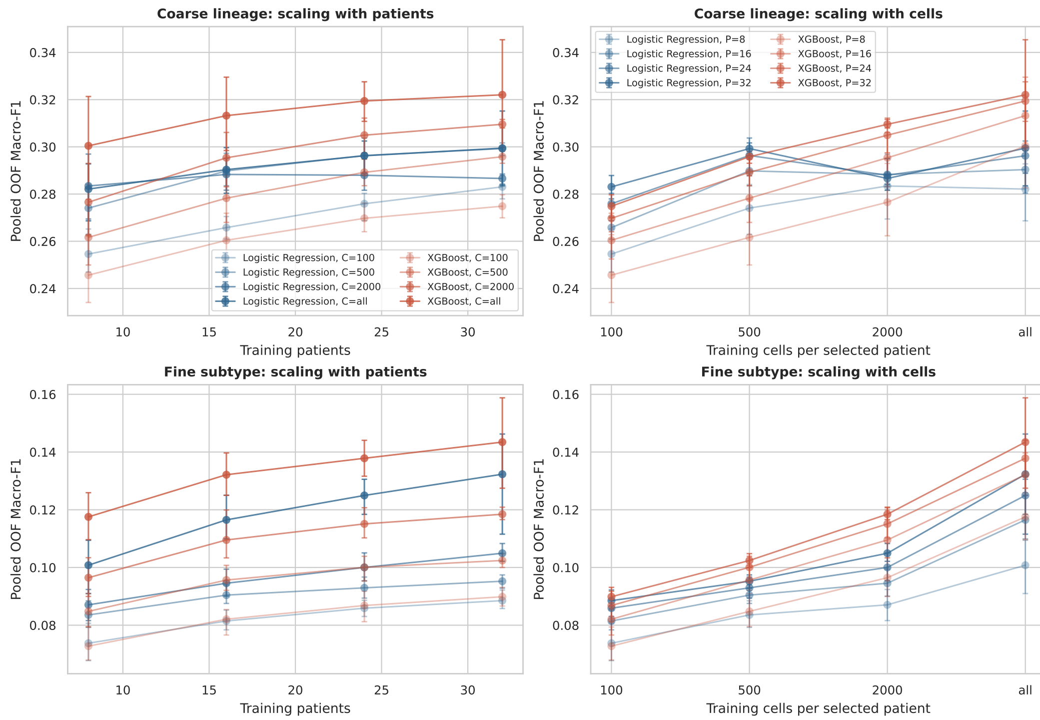

Figure 7: Patient-budget curves at each fixed per-patient cell cap. Intervals show subset-resampling variability; they do not represent uncertainty over a new patient population. Non-monotonic local changes are retained rather than smoothed.

The surface is cohort- and range-specific. We do not compare its unequal endpoints or use it to establish a patient- or cell-dominant scaling law, and it cannot estimate the benefit of patients drawn from new sites or acquisition batches.

C.16 Coarse–Fine Consistency Diagnostic

Independent coarse and fine classifiers can disagree about the hierarchy even when each prediction is plausible in isolation. Table 24 reports their natural agreement. Unlike an earlier single-split analysis, this version uses the same five-fold pooled out-of-fold prediction artifacts as the main hierarchy results, so every patient contributes exactly once to both evaluations.

Model	Fine Acc.	Root-excl. Hier. F1	Consistency
Soft-cascade Transformer	0.305	0.404	0.812
Flat FT-style Transformer	0.315	0.411	0.795
FT-style Transformer + HCE	0.306	0.405	0.834
XGBoost	0.323	0.420	0.736

Table 24: Auxiliary hierarchy diagnostics on the same 361,792-cell pooled OOF artifacts as Table 2. Root-excluded hierarchical F1 is the example-averaged F1 between the true and predicted $\{$lineage, subtype$\}$ sets; the always-correct root is omitted. Consistency maps each hard fine prediction to its parent and compares it with the independent coarse prediction. *Unknown* is excluded and no constrained decoding is used.

C.17 Feature-Redundancy Sensitivity of the Mixed Disease-Associated Signal

We asked how many of the 293 aptamers preserve the observed within-cohort disease-associated result. Because biological and technical sources are not identifiable, this section characterizes feature redundancy of a mixed signal; it does not validate a reduced assay, clinical deployment, or biomarker panel. Patient-Stable Aptamer Selection (PSAS) denotes the fold-specific ranking and development-only budget-selection protocol below; "stable" refers to comparison of ranks across outer folds, not to equal-patient-weight fitting of the LR ranker.

Algorithm 1: PSAS within outer fold o.

1. Seal the eight outer-test patients; denote the other 32 patients by D_o.
2. Using all cells in D_o, fit development-only standardization and an L1-regularized six-class multinomial LR (SAGA, $C = 0.1$); rank aptamer j by $s_j = \sum_{d=1}^{6} |\beta_{dj}|$.
3. Seed-shuffle the patients in D_o into five patient-disjoint, non-stratified inner folds.
4. For every $B \in \{5, 10, 20, 40, 80, 160, 293\}$ and inner fold, refit LR on the inner-training patients using the fixed Top-B ranking and score patient-majority-vote macro-F1 on inner-validation patients.
5. Let $\bar{F}_B$ denote mean inner-validation macro-F1 and select $B^* = \min\{B : \bar{F}_B \geq \bar{F}_{293} - \delta\}$ with $\delta = 0.02$.
6. Refit LR and XGBoost from scratch on all patients in D_o, using only the LR-selected Top-B^* features, and evaluate once on the sealed outer-test patients.
7. Repeat independently for all five outer folds; report the pooled outer-test predictions.

Implementation scope and controls. The LR ranker is a cell-level disease model: cells are not first aggregated to patient medians and receive equal weight, so its coefficients are cell-count weighted rather than equal-patient-weighted. The full-panel reference $\bar{F}_{293}$ is the mean inner-validation score, not a resubstitution estimate on all of D_o. The absolute margin $\delta = 0.02$ was fixed in code before outer-test scoring but was not prospectively preregistered. Because the ranking is computed once from all of D_o and then held fixed across its inner folds, budget selection is development-only but conditional on that ranking, not fully nested inner re-ranking.

For Random-B, each $B \in \{5, 10, 20\}$ has 100 uniformly sampled feature sets. The feature set for one repeat is shared across the five outer folds, but LR and XGBoost are trained from scratch within every fold. XGBoost in Table 25 and Figure 8 uses the LR-selected panel. The separate SHAP control instead fits full-panel XGBoost in each outer-development set, constructs an equal-patient-weight mean absolute TreeSHAP ranking, and refits both classifiers on SHAP-selected Top-10/20 features; it does not determine B^* or the main curve.

All five outer folds select $B^* = 10$ under this rule.

Budget terminology. An *aptamer measurement budget* B means that a model is retrained and evaluated using only those B input features. An *inference-time cell budget* C means that, after training on all available outer-development cells, only C cached predictions are sampled from each held-out patient before majority-vote aggregation. We do not reduce or evaluate a *training-time cell budget*; the cell-count sensitivity therefore supports low-cell inference aggregation, not training with only C cells per patient.

The resulting development-selected 10-aptamer panels reproduce the full-panel patient-level macro-F1 of 1.0 for both Logistic Regression and XGBoost on all five outer test folds. Figure 8 compares these selected panels with 100 Random-B repeats at each compact budget. For every model and budget, we report the random-panel mean, empirical 2.5th–97.5th percentile interval, and the empirical midrank percentile of Top-B within that distribution (Table 26).

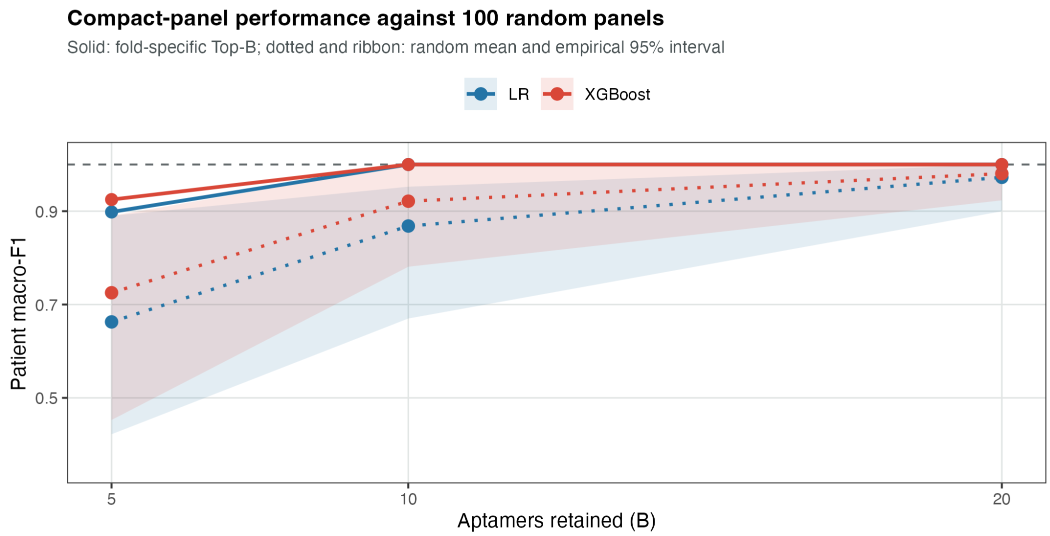

Figure 8: Patient-level disease-association macro-F1 under feature-budget controls. Solid lines show pooled out-of-fold performance obtained using fold-specific Top-B panels; dotted lines and ribbons show the mean and empirical 95% interval over 100 random-panel repeats. Each repeat shares one panel across outer folds, with models refit foldwise. The dashed line is the full 293-aptamer reference; scores pool the 40 outer-test patients once each. Because acquisition effects are unrecorded, the curves quantify redundancy of the mixed within-cohort signal rather than reduced-assay validity.

Outer Fold	δ	B^*	LR Macro-F1	XGBoost Macro-F1
0	0.02	10	1.000	1.000
1	0.02	10	1.000	1.000
2	0.02	10	1.000	1.000
3	0.02	10	1.000	1.000
4	0.02	10	1.000	1.000
All folds	**0.02**	**10**	**1.000**	**1.000**

Table 25: Outer-test-isolated feature-budget sensitivity for the mixed disease-associated signal. In each outer fold, aptamer ranking and budget selection are performed using outer-development patients only, and the selected budget is evaluated once on the untouched outer-test patients. The ranking is fixed across inner validation splits; B is selected conditional on that ranking. Under the analysis-specified, non-preregistered tolerance margin $\delta = 0.02$, all five folds select the same development budget $B^* = 10$, and the resulting 10-aptamer panels preserve the observed full-panel patient macro-F1 for both Logistic Regression and XGBoost. This result demonstrates feature redundancy of the observed mixed disease-associated signal. It does not validate a reduced assay, a biological signature, or unique causal biomarkers.

B	Model	Top-B	Random mean [95% interval]	Percentile
5	LR	0.898	0.663 [0.422, 0.890]	97.0
5	XGBoost	0.925	0.725 [0.453, 0.890]	99.0
10	LR	1.000	0.868 [0.670, 0.952]	100.0
10	XGBoost	1.000	0.921 [0.781, 1.000]	97.0
20	LR	1.000	0.973 [0.900, 1.000]	79.5
20	XGBoost	1.000	0.981 [0.923, 1.000]	75.0

Table 26: Feature-redundancy controls over 100 panel repeats. Top-B uses fold-specific outer-development LR rankings. Each repeat draws one random panel shared across outer folds; both classifiers are refitted in every fold. The percentile is the empirical midrank of Top-B within the random distribution. Because disease and unrecorded acquisition effects are not identifiable, these values characterize only the mixed within-cohort signal.

At $B = 10$, the Top-B panel is at the 100th empirical percentile for LR and the 97th for XGBoost; Random-B means are 0.868 and 0.921, respectively. By $B = 20$, random means rise to 0.973 and 0.981 and many random panels also saturate, reducing the corresponding midrank percentiles to 79.5 and 75.0. Thus ranking is most consequential at the smallest budgets, while the broader panel contains extensive substitutable signal.

We next crossed aptamer budget $B \in \{5, 10, 20, 293\}$ with inference-time cell caps $C \in \{100, 500, 1000, 5000, \text{all}\}$. For each finite cap, cached predictions were sampled without replacement within each outer-test patient; patient majority votes were pooled across the five held-out folds, and the process was repeated over 50 sampling seeds. Training still used all outer-development cells. At $B = 5$, macro-F1 varies with the sampled cells and reaches 0.881 on average at $C = 100$; at $B = 10$, 20, and 293, macro-F1 is 1.0 for every cap and every seed (Table 27).

B	100 cells	500 cells	1,000 cells	5,000 cells	All cells
5	0.881 [0.845, 0.925]	0.894 [0.852, 0.925]	0.906 [0.875, 0.925]	0.899 [0.898, 0.898]	0.898 [0.898, 0.898]
10	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
293	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]

Table 27: LR patient-level macro-F1 under inference-time equal-cell subsampling. Entries are mean [empirical 95% interval] over 50 seeds; cached outer-test predictions are subsampled before aggregation. Models are trained once using all outer-development cells, so this is not a training-time cell-budget experiment.

Finally, we tested whether the compact result depends on Logistic Regression coefficients as the attribution mechanism. Within each outer-development set, we fitted a full-panel XGBoost model and computed absolute TreeSHAP contributions on at most 1,000 cells per development patient. We first averaged over cells within each patient and then over patients, preventing high-cell-count patients from dominating the attribution ranking. The LR–SHAP Jaccard overlap is moderate rather than complete (0.501 for Top-10 and 0.475 for Top-20), yet panels from either ranking achieve patient macro-F1 1.0 when evaluated with either LR or XGBoost (Table 28). This cross-model control supports predictive redundancy of the mixed signal, not measurement efficiency, molecular specificity, or a unique signature.

B	LR–SHAP Jaccard	Ranking	LR eval.	XGB eval.
10	0.501 ± 0.126	LR coefficient	1.000	1.000
10	–	XGBoost-SHAP	1.000	1.000
20	0.475 ± 0.087	LR coefficient	1.000	1.000
20	–	XGBoost-SHAP	1.000	1.000

Table 28: Cross-model attribution control. Jaccard is mean $\pm$ SD across five outer-fold Top-B sets. LR-coefficient and XGBoost-SHAP rows use their own selected features; evaluation columns report pooled patient macro-F1 after refitting LR or XGBoost. SHAP ranking does not select B^*.

The fold-overlap LR ranking is also stable. The highest-ranked aptamers by mean outer-fold rank are APT-179, APT-225, APT-164, APT-166, APT-57, APT-69, APT-125, APT-161, APT-172, and APT-218, and 290 of the 293 aptamers are retained after a simple redundancy screen at $|r| \leq 0.9$. These anonymized IDs have no verified public protein-target mapping, and feature ranking under unresolved confounding cannot support protein-target, pathway, or molecular-plausibility interpretation. We therefore omit the earlier full-cohort attribution visualization.

C.18 BALANCED SAMPLING ABLATION

A development-only class-balanced-sampling variant did not outperform natural sampling after matching optimization steps and was not retained.

C.19 EXPLORATORY LEAVE-ONE-DISEASE-OUT STRESS TEST

We retain only a leave-one-disease-out (LODO) representation stress test. For each run, all patients from one disease category are withheld from training and used only for evaluation. Table 29 shows that generalization weakens substantially relative to the standard patient-disjoint folds, especially for fine subtypes. Because there are only six disease groups and no external cohort, LODO is descriptive rather than an estimate of deployment robustness.

Held-out Disease	Fine Macro-F1	Fine Acc.	Coarse Macro-F1	Coarse Acc.	Tail Macro-F1
BTC	0.086	0.317	0.310	0.479	0.073
CRC	0.105	0.277	0.309	0.520	0.112
GC	0.063	0.269	0.281	0.488	0.057
HCC	0.063	0.246	0.264	0.383	0.072
PC	0.057	0.253	0.221	0.482	0.037
Macro-average	**0.075**	**0.272**	**0.277**	**0.470**	**0.070**

Table 29: Leave-one-disease-out generalization for Soft-cascade Transformer over the five malignant held-out diseases. Each row withholds all patients of the named disease from training. The in-distribution reference using the same model under the standard patient-disjoint 5-fold protocol is fine Macro-F1 0.152 and coarse Macro-F1 0.326. A separate Normal-held-out run produces near-chance fine performance. LODO is reported only as an exploratory appendix stress test, not as proof that the model already solves cross-disease generalization.

We omit unmatched detector rankings. Cross-disease generalization remains unresolved and requires external cohorts with prespecified distribution shifts and patient-clustered uncertainty.